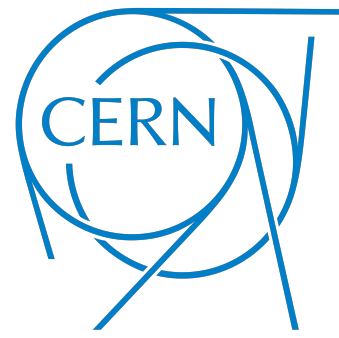
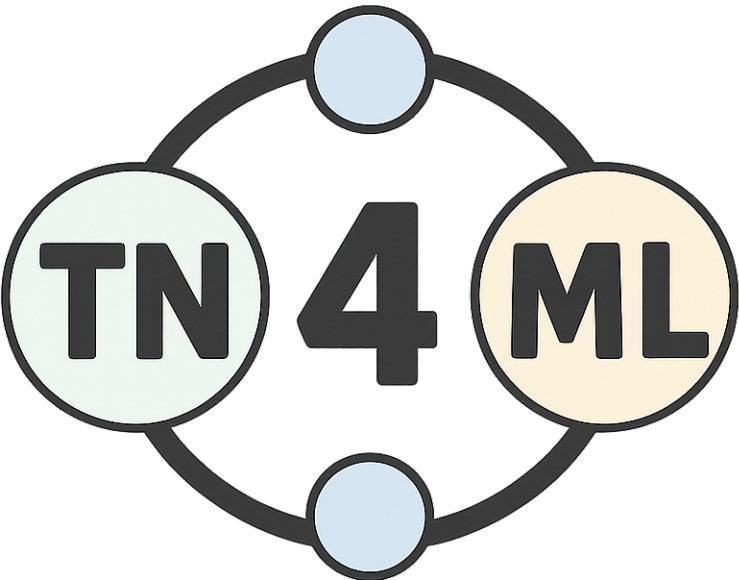




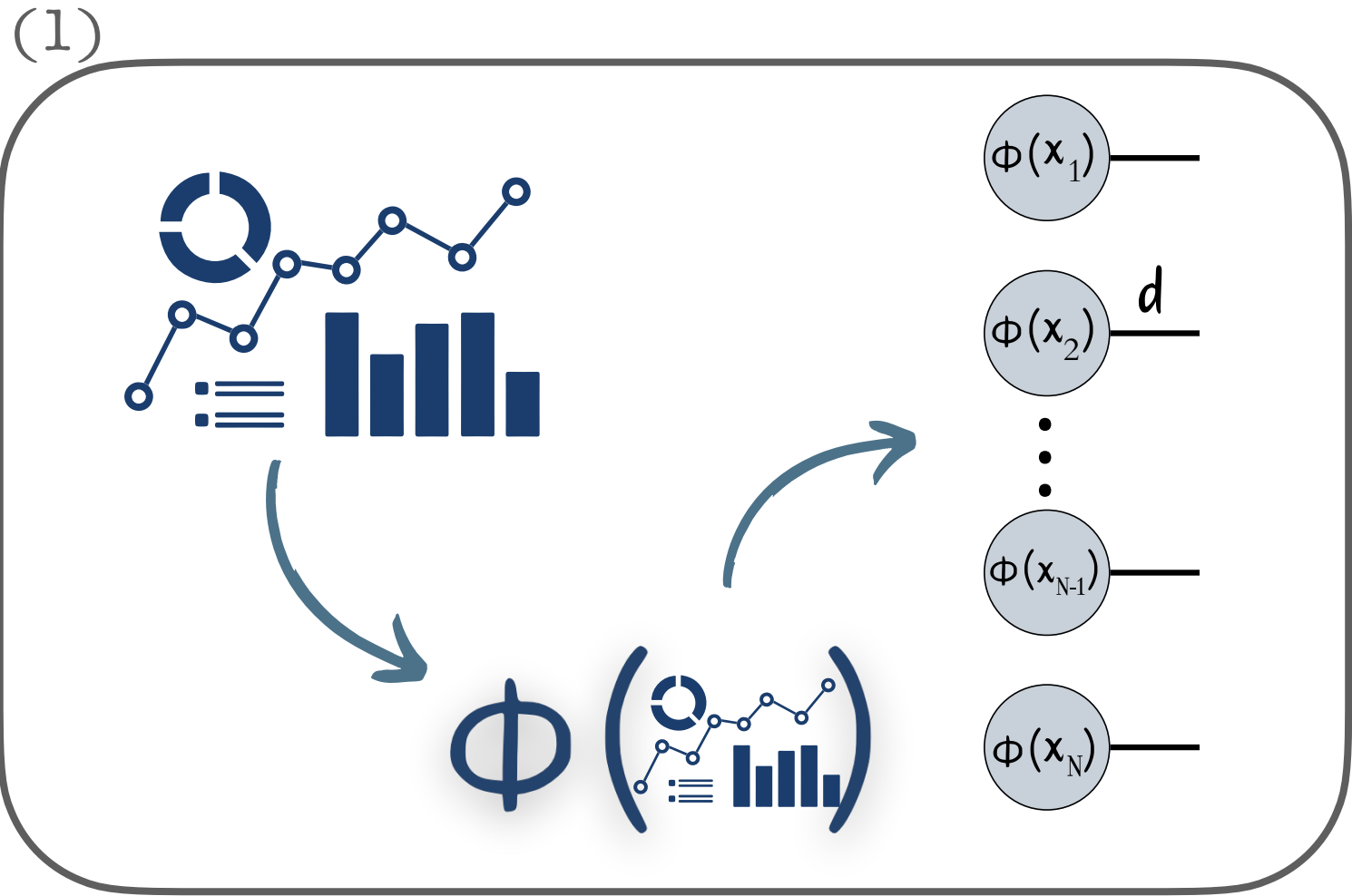
Tensor Network for Anomaly Detection in the Latent Space of Proton Collision Events at the LHC

Ema Puljak, Maurizio Pierini, Artur Garcia Saez

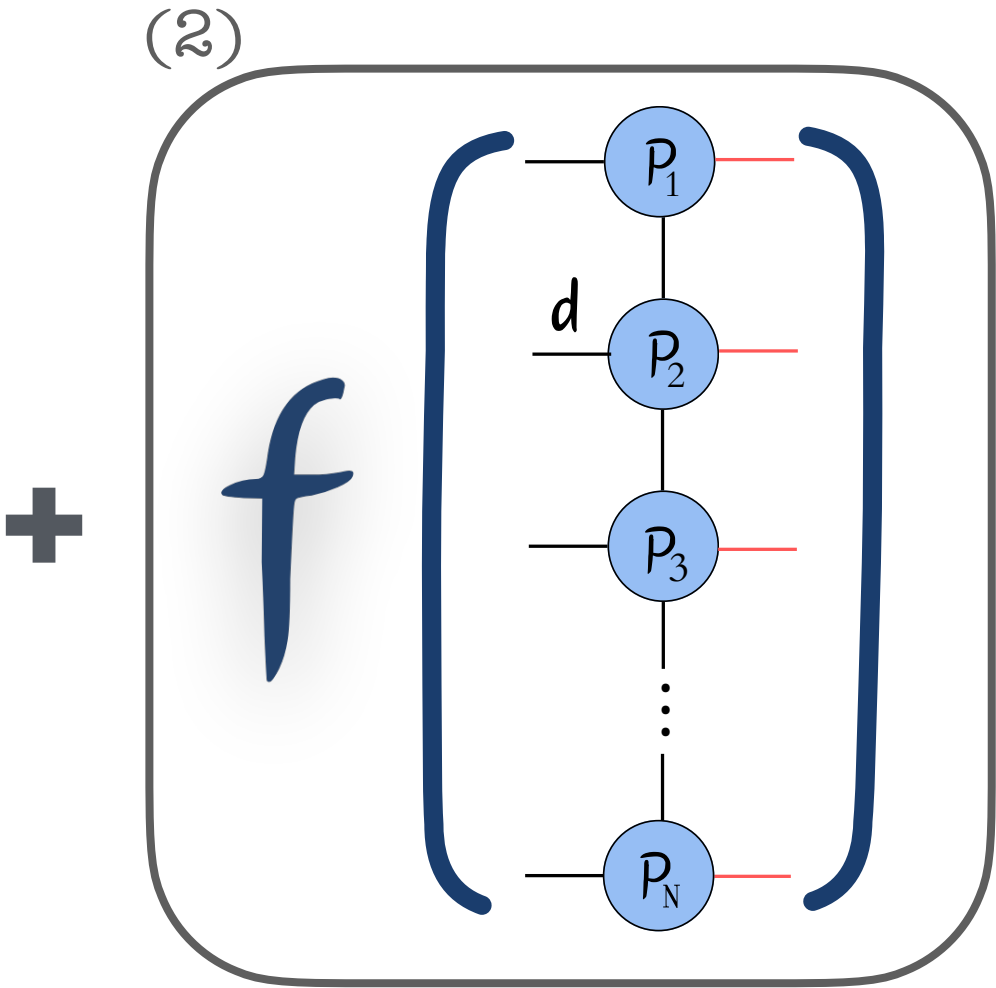
Tensor Networks for Machine Learning



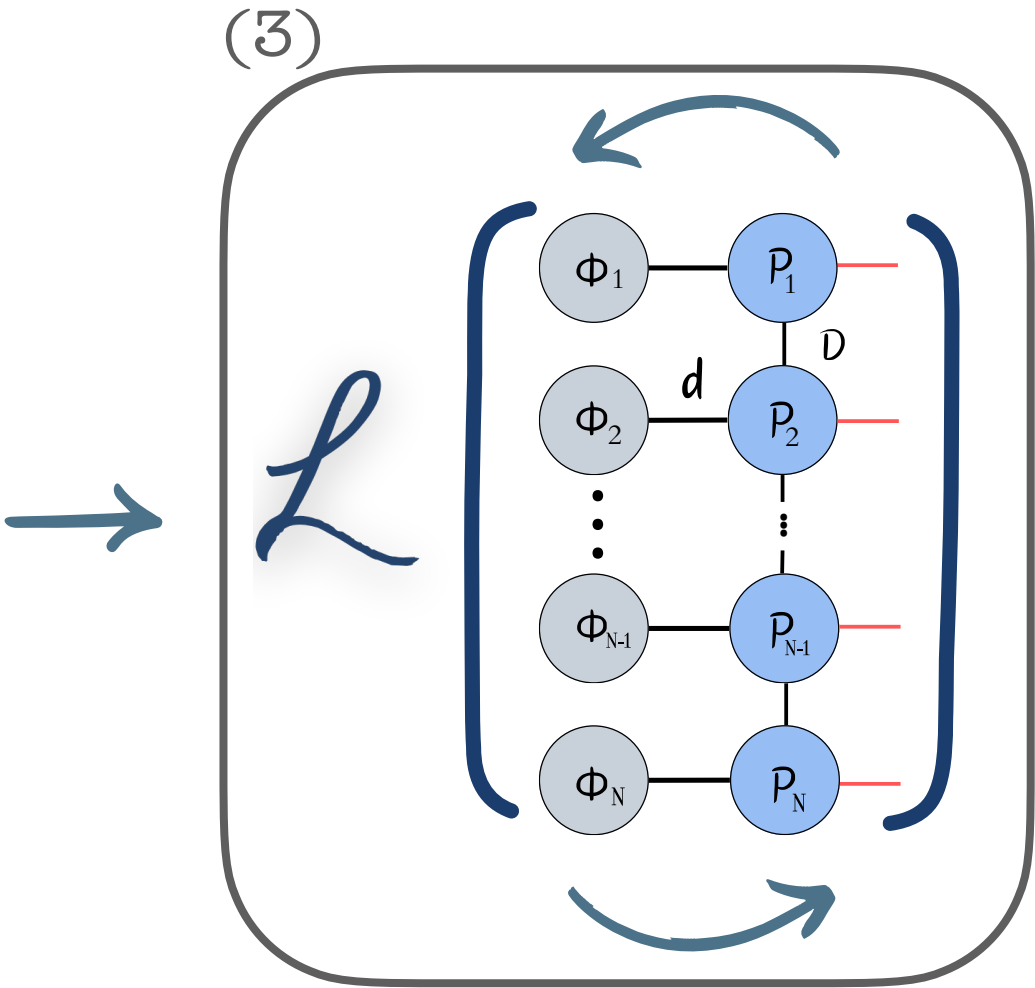
[\[github.com/bsc-quantum/tn4ml\]](https://github.com/bsc-quantum/tn4ml)



Data Embedding



Initialization



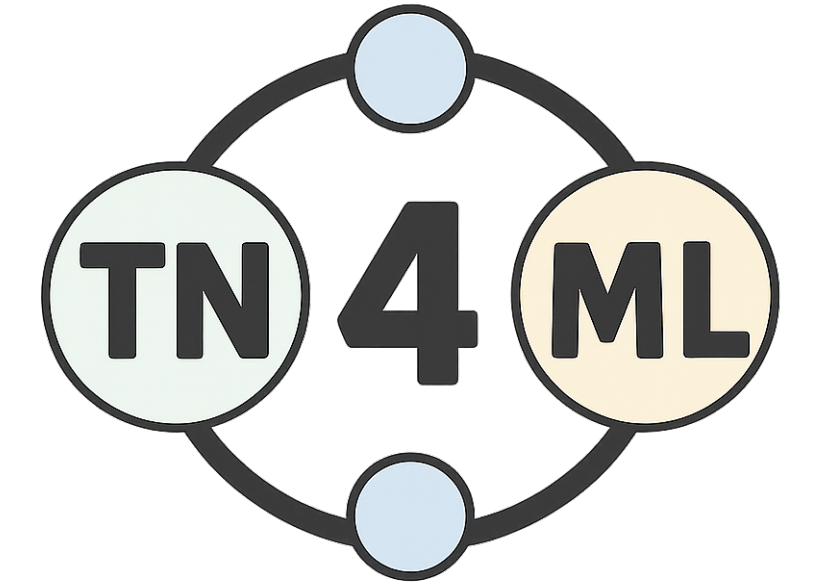
Optimization



Evaluation

QUIMB $\longrightarrow$ Tensor Networks

$\longrightarrow$ Optimization



Applications

HEP data

TNs for Anomaly Detection in the Latent Space of Proton Collision Events at the LHC

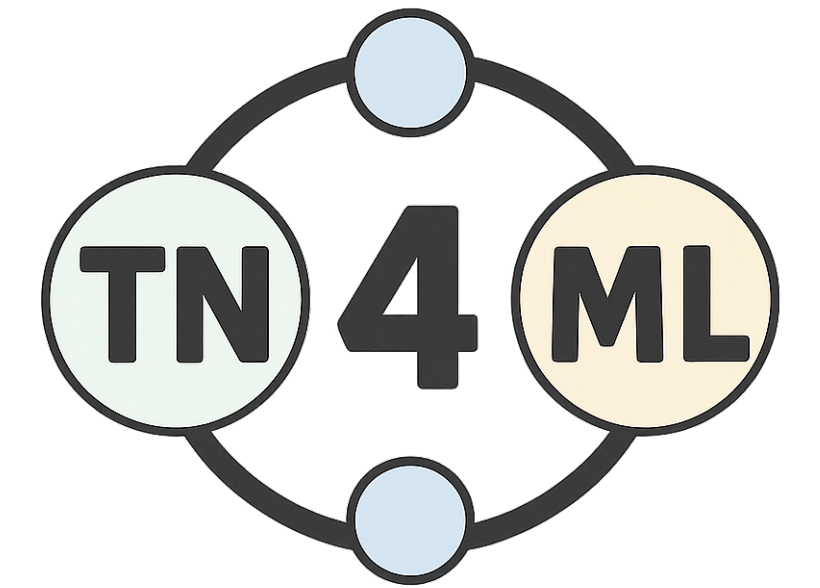
Towards Ultrafast TN Models for Classification on FPGAs

Medical data

TNs for Cancer Detection in Medical Imaging

TNs for Effective Data Embeddings and Medical Image Classification

Breast Cancer Classification with Tensor Networks



Applications

HEP data

TNs for Anomaly Detection in the Latent Space of Proton Collision Events at the LHC

Towards Ultrafast TN Models for Classification on FPGAs

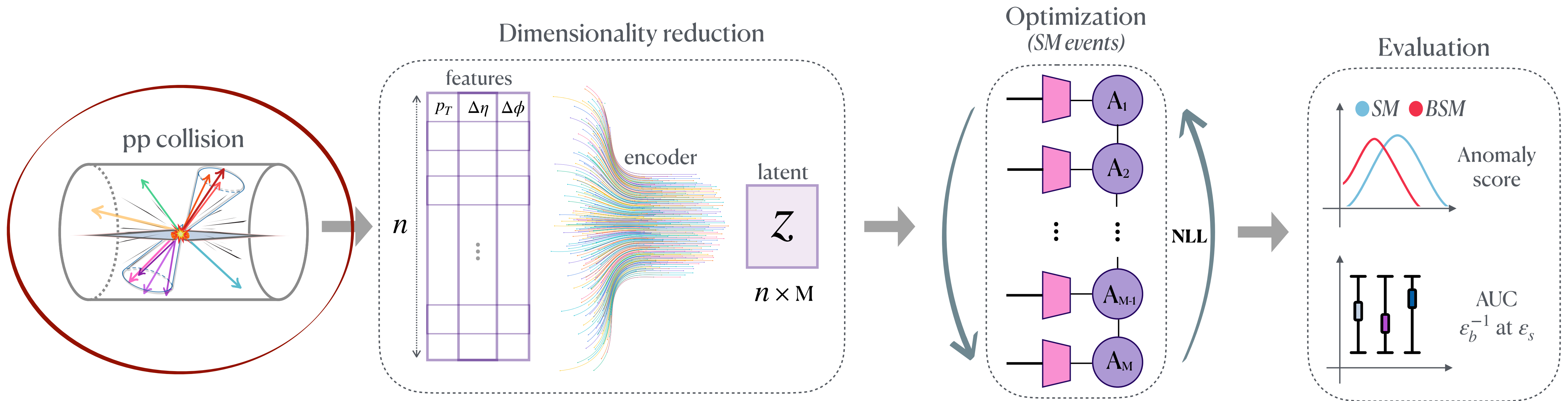
Medical data

TNs for Cancer Detection in Medical Imaging

TNs for Effective Data Embeddings and Medical Image Classification

Breast Cancer Classification with Tensor Networks

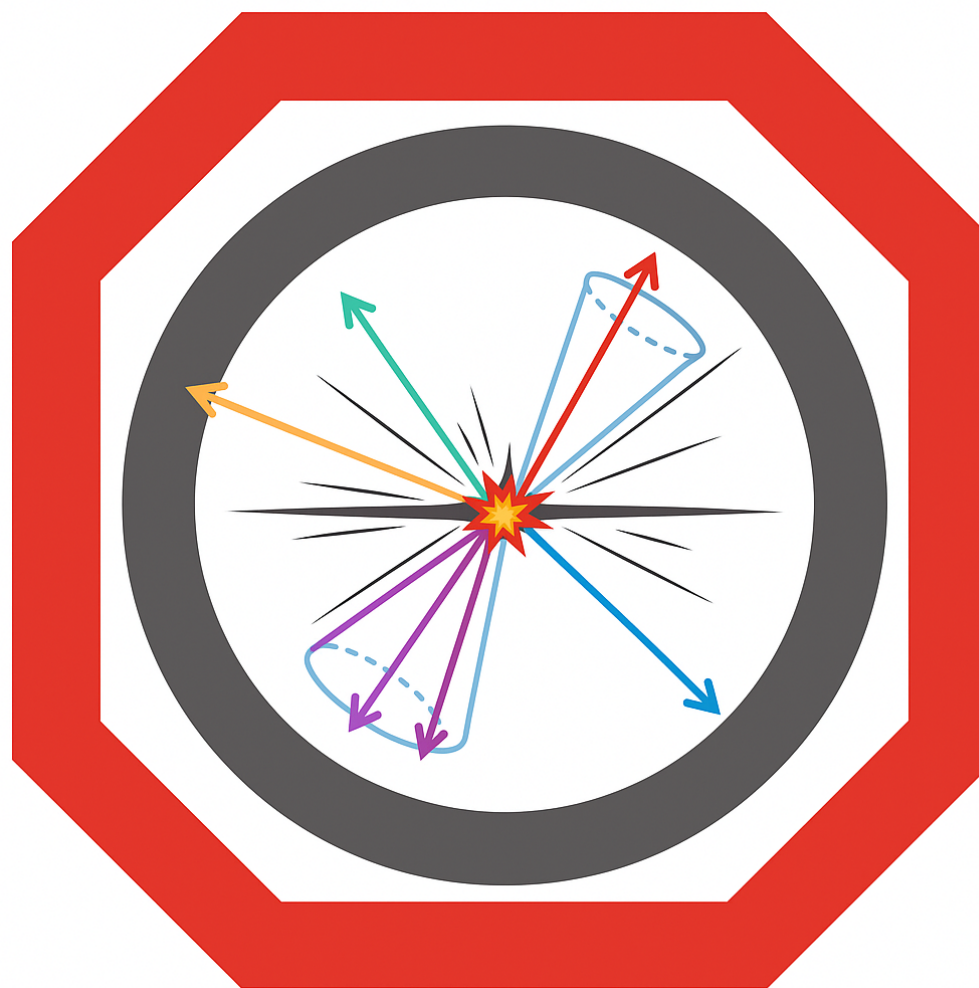
TNs for Anomaly Detection in the Latent Space of Proton Collision Events at the LHC



TNs for Anomaly Detection in the Latent Space of Proton Collision Events at the LHC



Detector

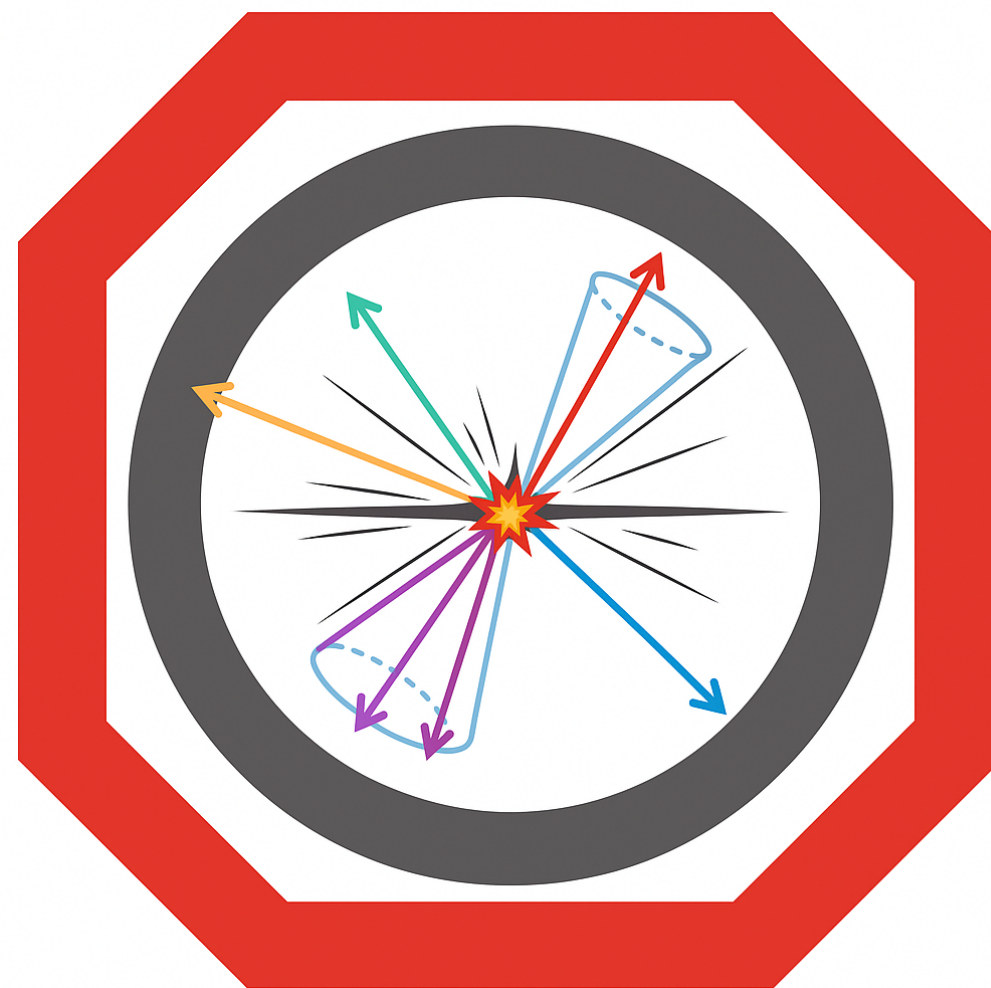


40 MHz collision rate

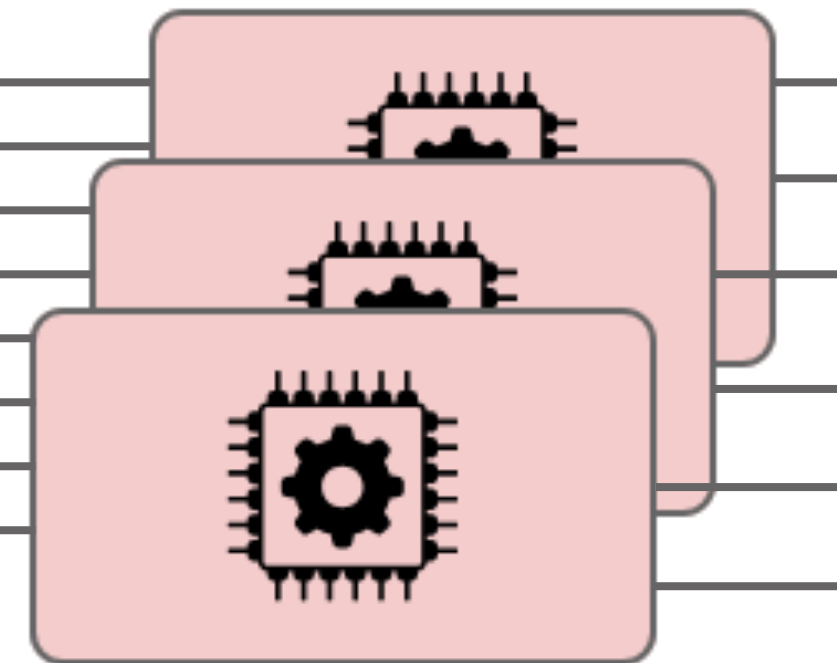
~30 TB/s of raw data

Two-level trigger reduces data rate
for offline analysis

Detector



L1 Trigger



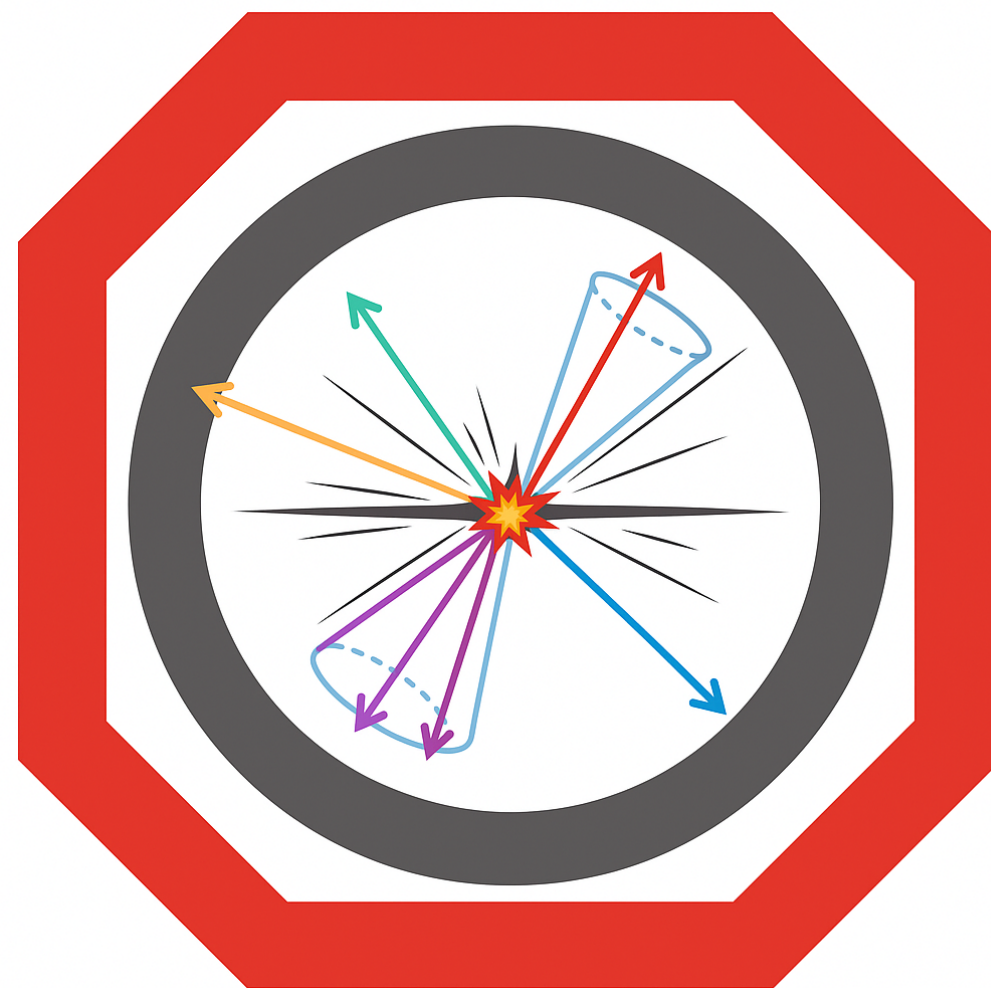
$\sim 99.75\%$ reject

40 MHz $\rightarrow \mathcal{O}(100)$ kHz

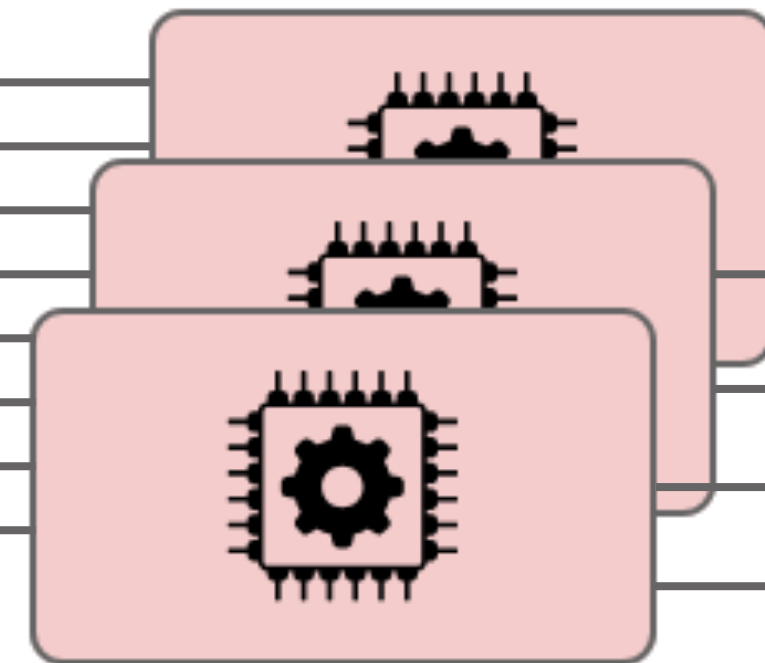
Hardware-based

$\mathcal{O}(1)\mu s$ latency decision

Detector

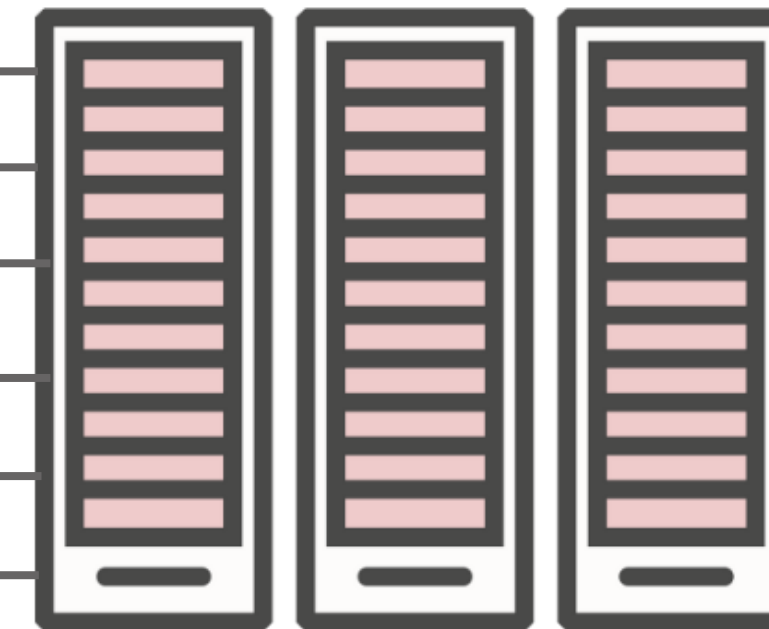


L1 Trigger



~ 99.75% reject

HLT Trigger

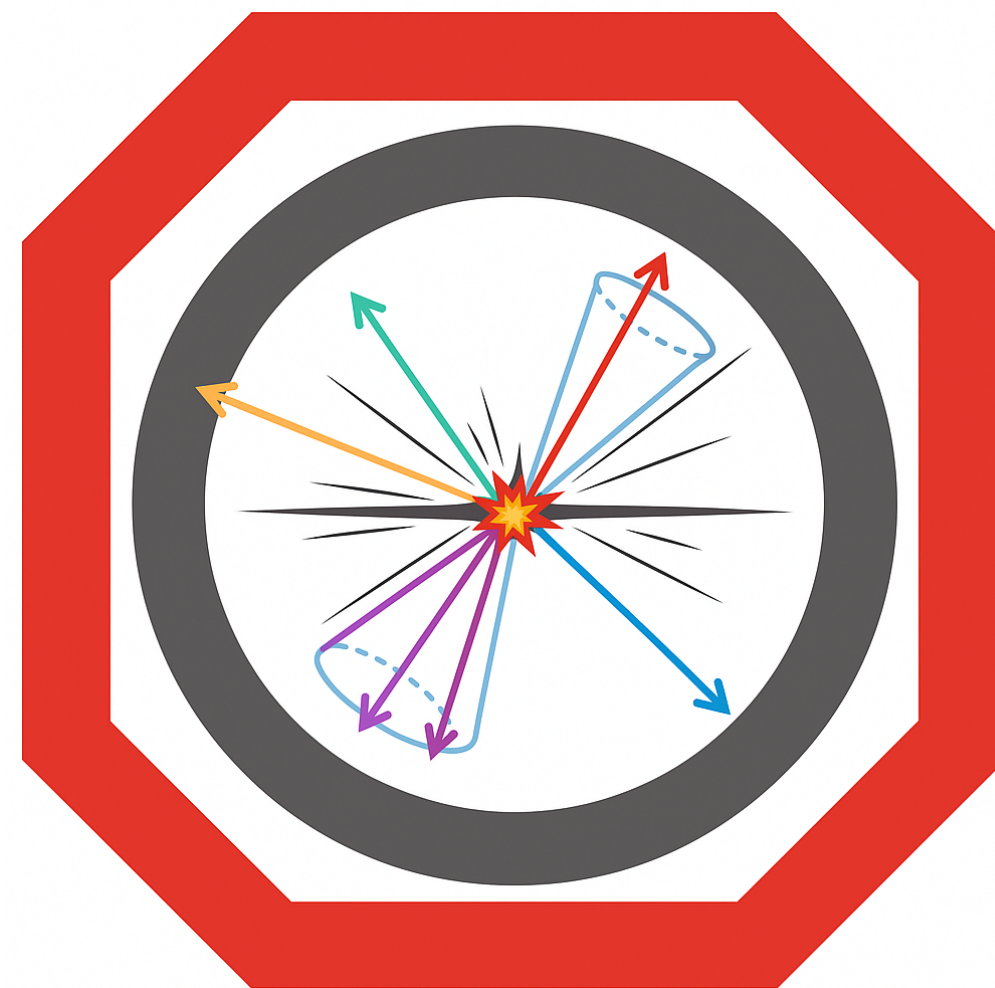


~ 99% reject

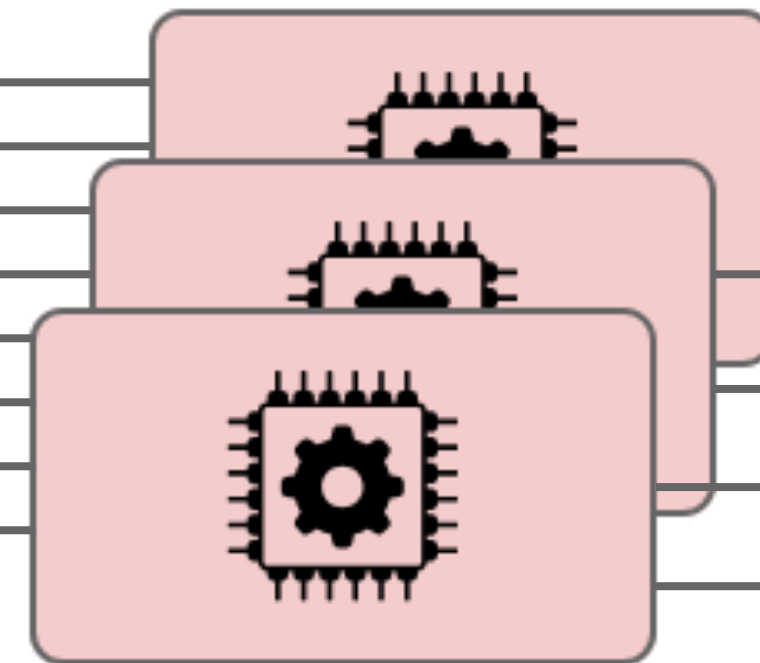
$\mathcal{O}(100) \text{ kHz} \rightarrow 1 \text{ kHz}$

Software-based trigger

Detector

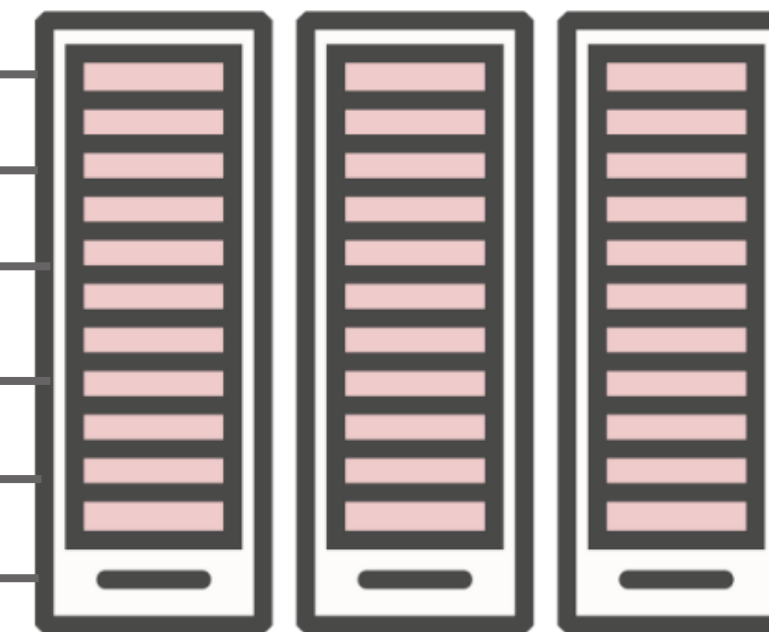


L1 Trigger



~ 99.75% reject

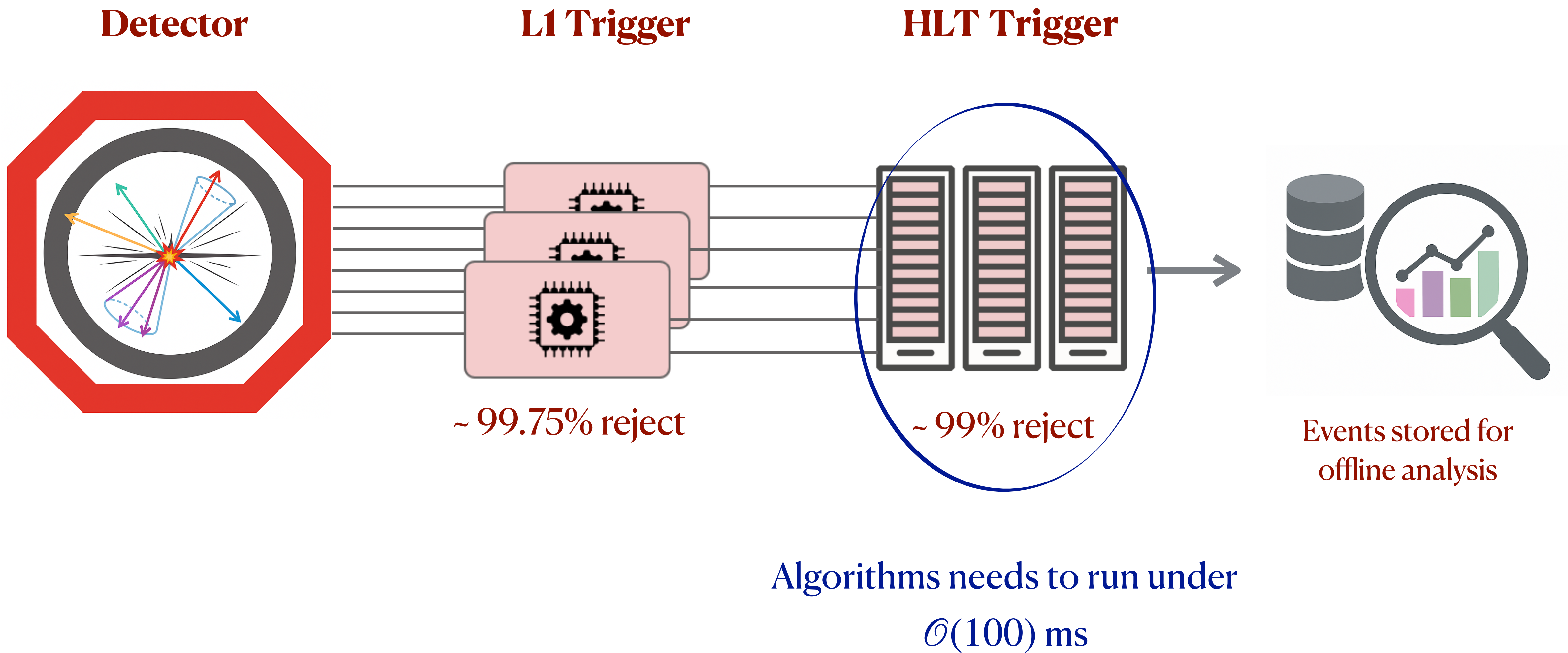
HLT Trigger



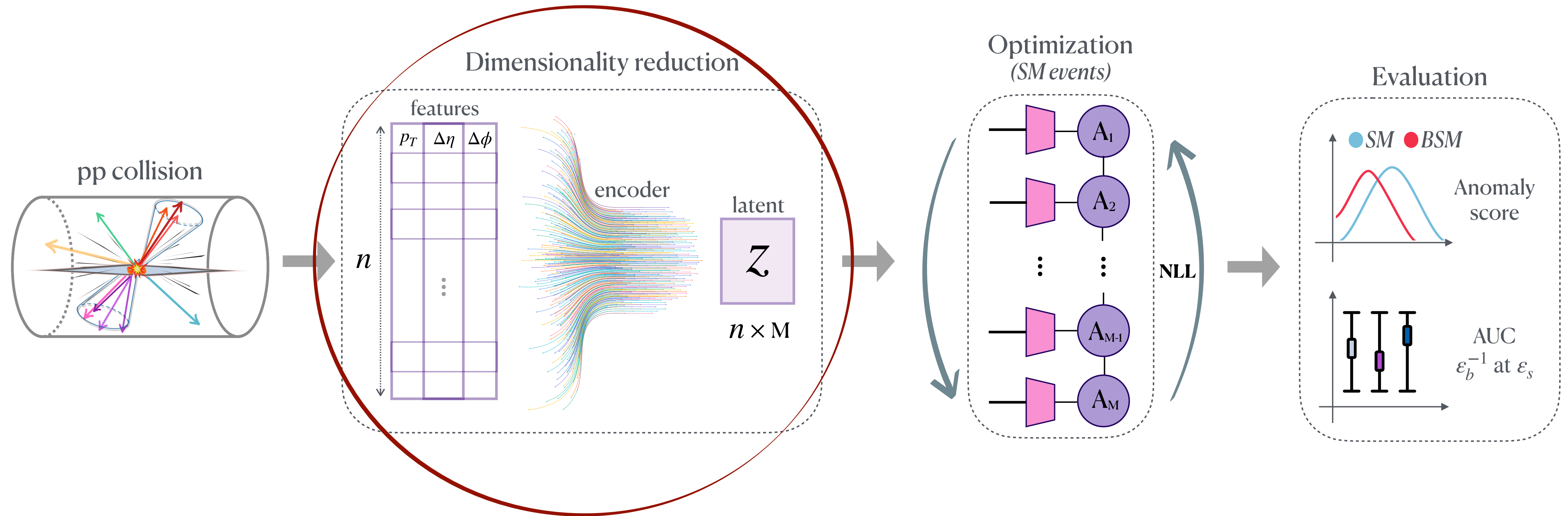
~ 99% reject



**Events stored for
offline analysis**

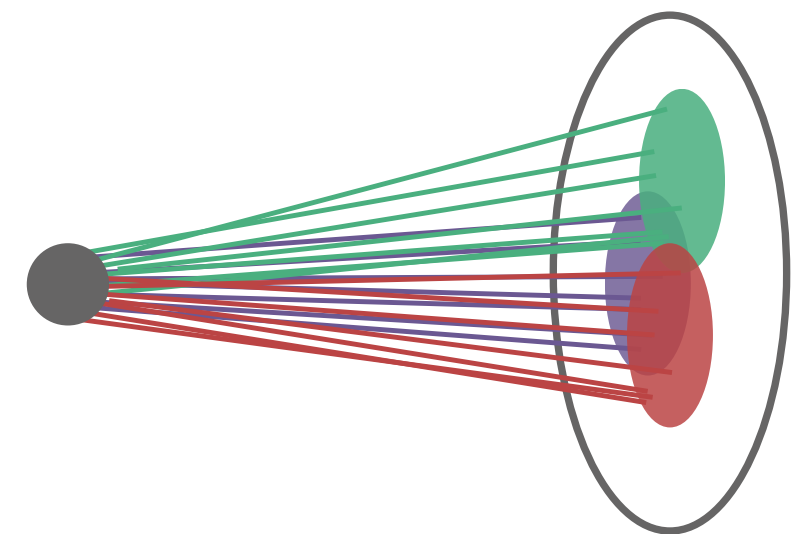
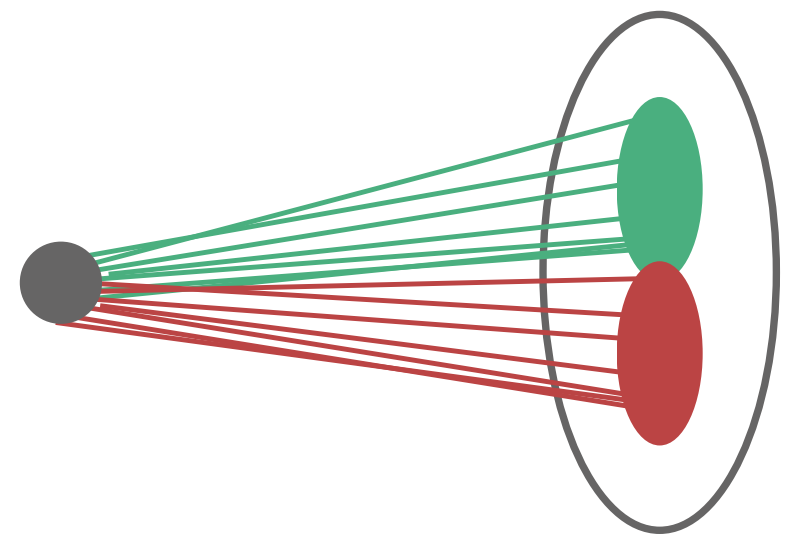
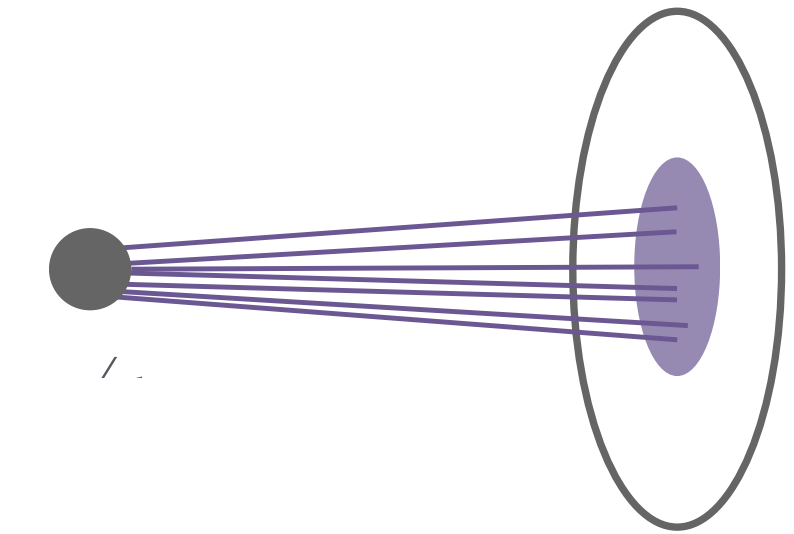


TNs for Anomaly Detection in the Latent Space of Proton Collision Events at the LHC



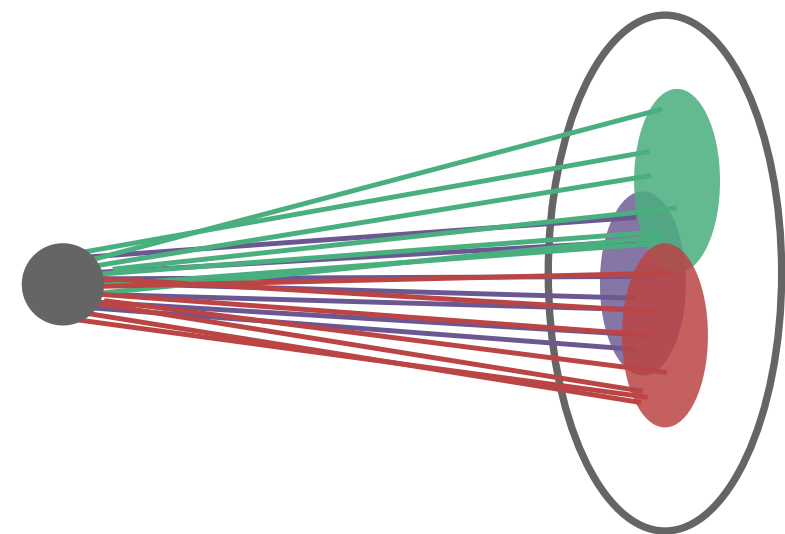
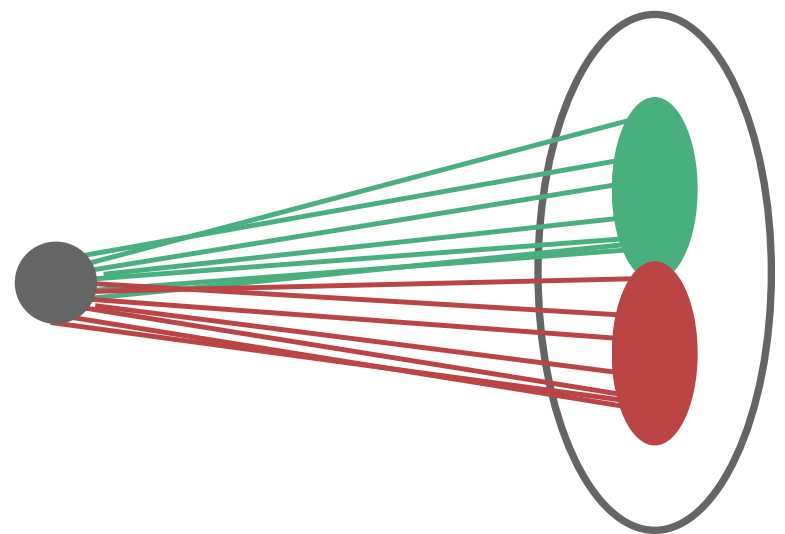
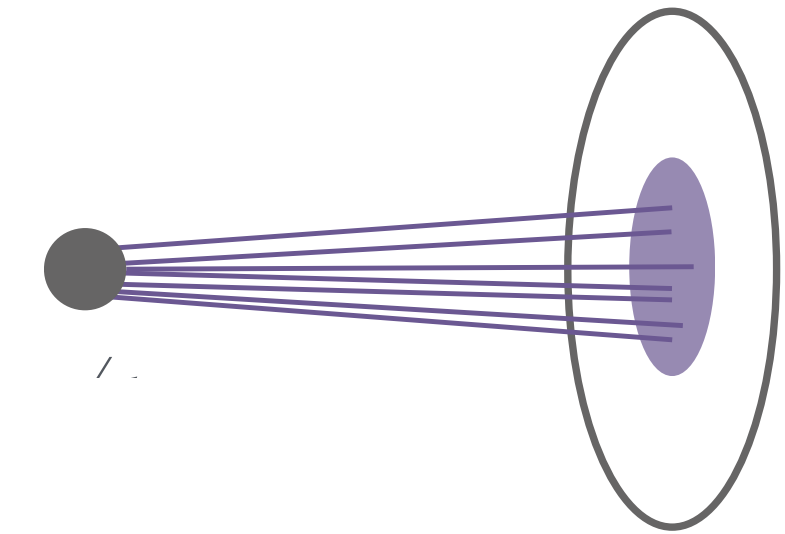
Dataset & Dimensionality Reduction

Jet = spray of particles



Dataset & Dimensionality Reduction

Jet = spray of particles

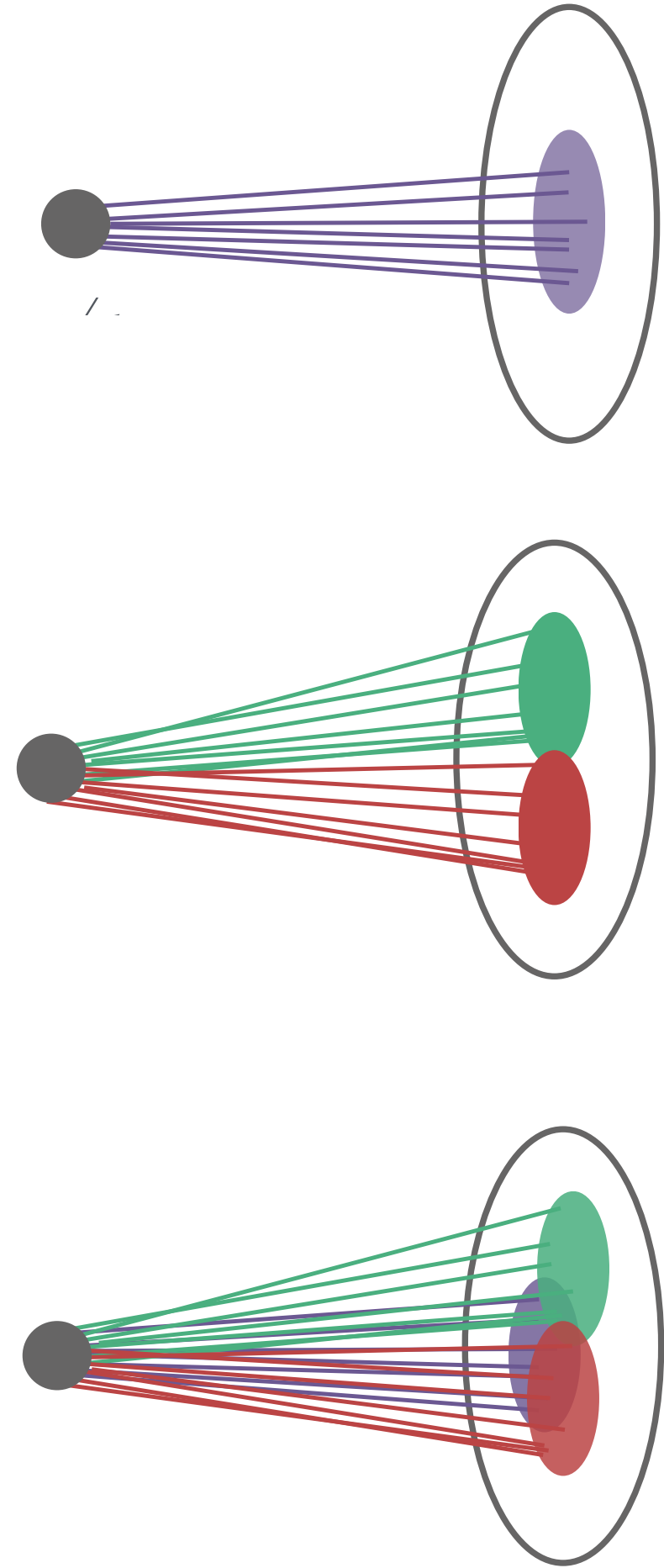


Chosen features

n	p_T	$\Delta\eta$	$\Delta\phi$
		$\vdots$	

Dataset & Dimensionality Reduction

Jet = spray of particles

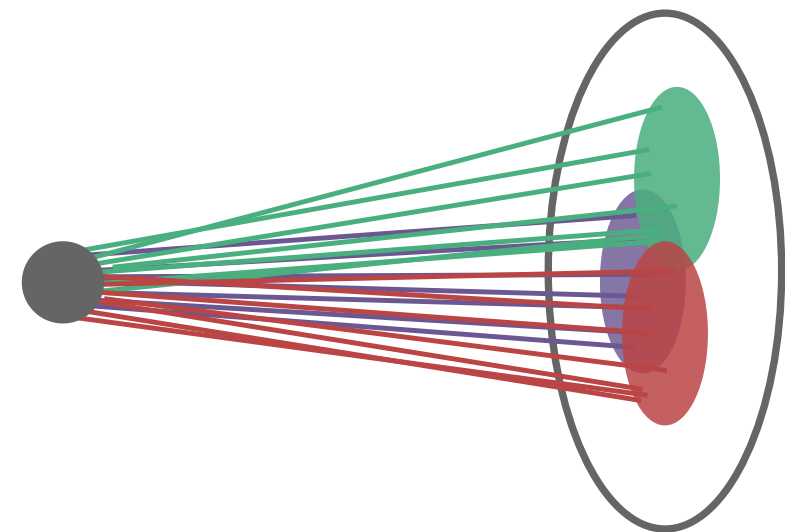
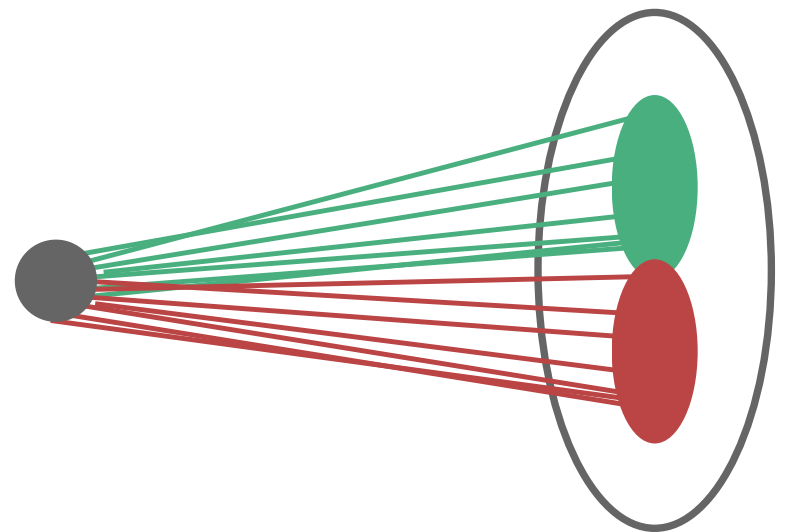
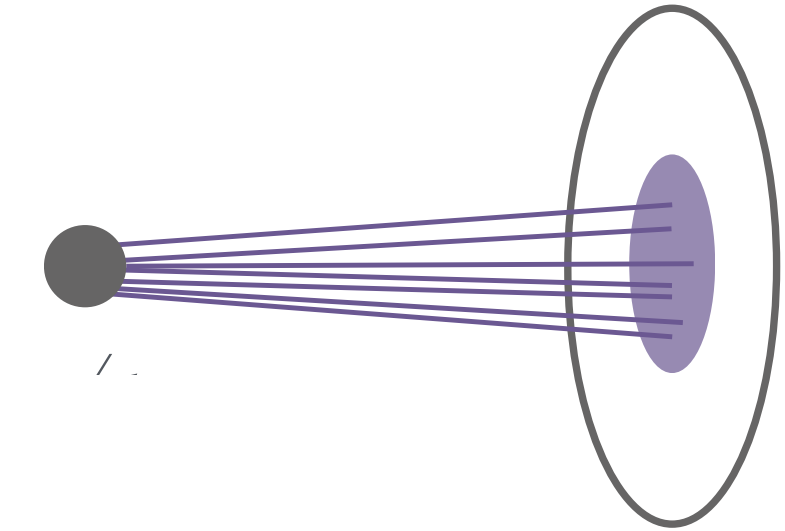


Chosen features

Diagram illustrating a 2D histogram structure with 100 rows and 3 columns. The columns are labeled p_T , $\Delta\eta$, and $\Delta\phi$. A vertical double-headed arrow on the left indicates the 100 rows. The center of the grid contains three vertically aligned dots, indicating the continuation of the grid.

Dataset & Dimensionality Reduction

Jet = spray of particles



Total = 300 features

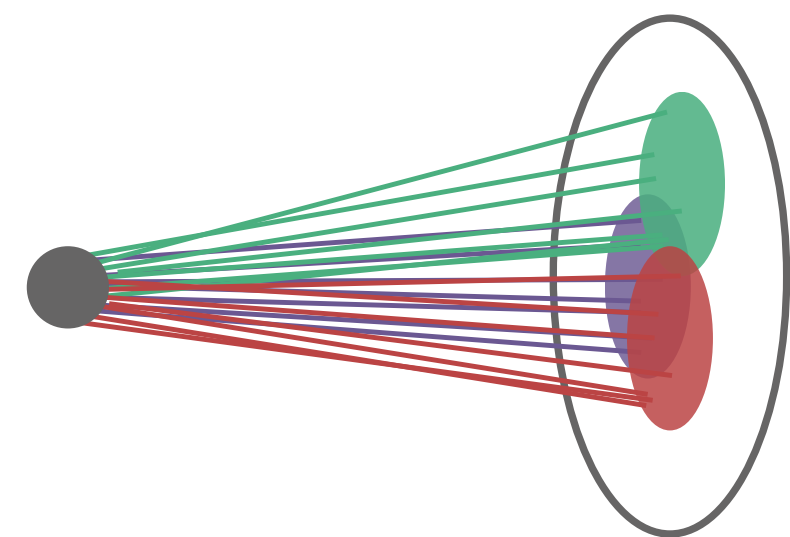
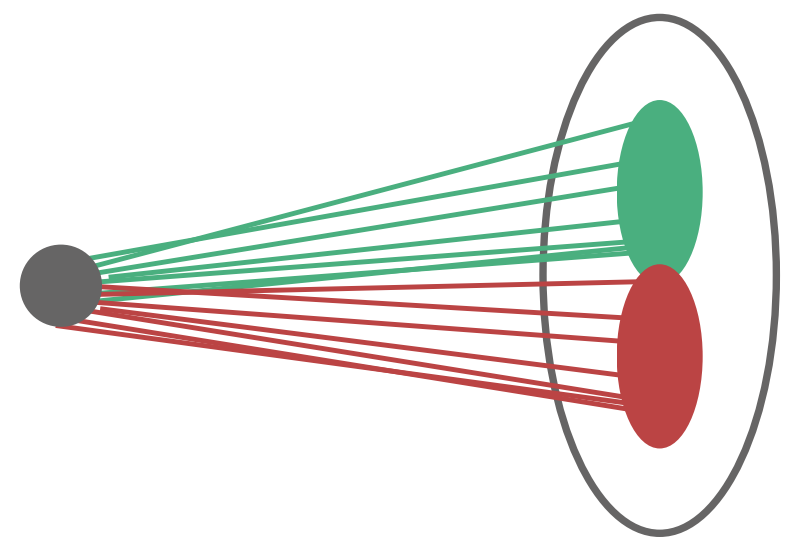
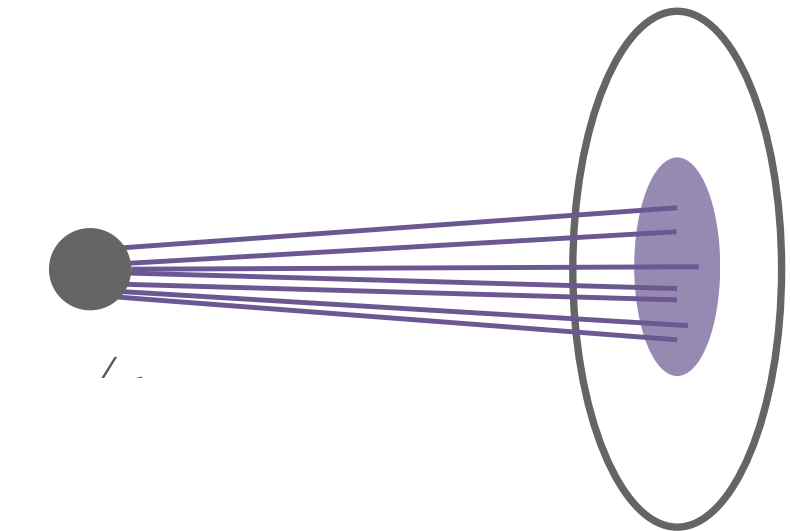


100

A diagram illustrating a 2D grid structure. The grid has three columns labeled p_T , $\Delta\eta$, and $\Delta\phi$. A vertical dashed arrow on the left points upwards and is labeled with a superscripted p .

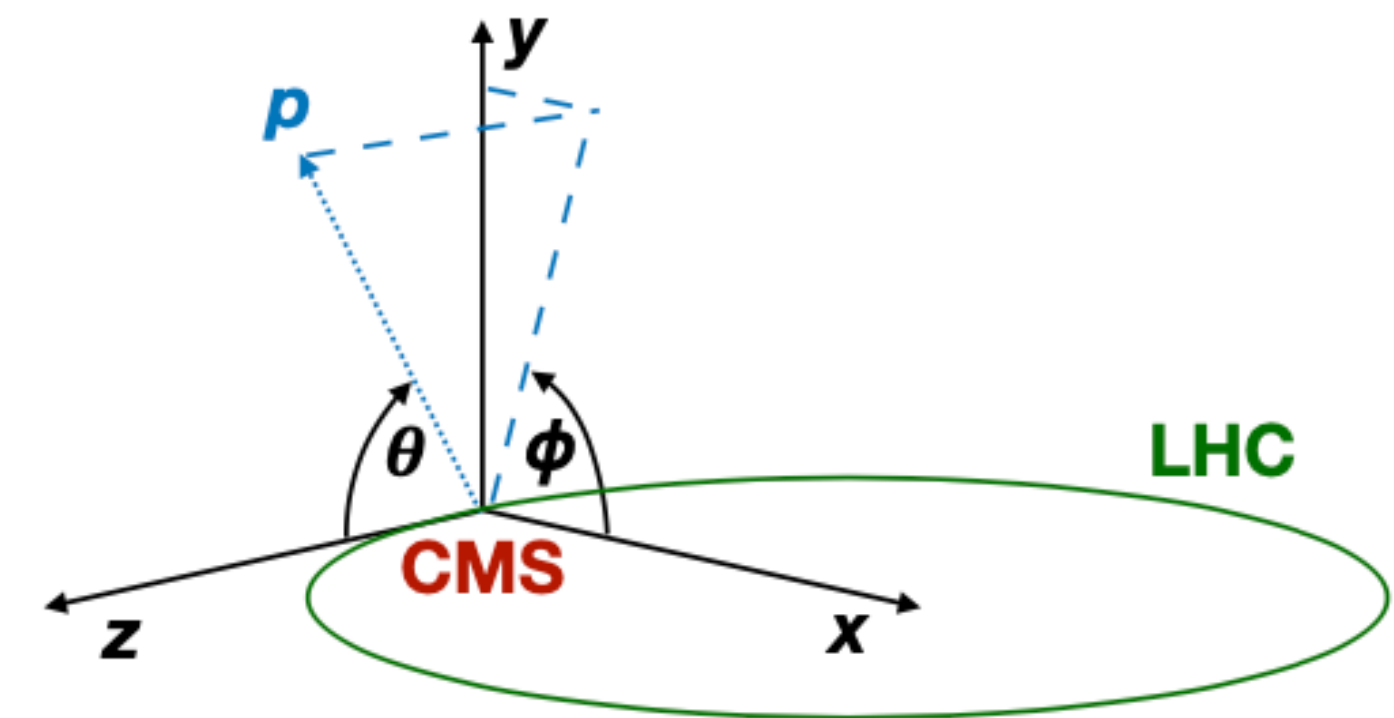
Dataset & Dimensionality Reduction

Jet = spray of particles



Total = 300 features

Diagram illustrating a 2D histogram structure with 100 rows and 3 columns. The columns are labeled p_T , $\Delta\eta$, and $\Delta\phi$. A vertical double-headed arrow on the left indicates the height of the grid is 100. The grid is divided into 100 rows and 3 columns by purple lines. The first row is labeled with p_T , $\Delta\eta$, and $\Delta\phi$. The middle row contains three vertical dots. The bottom row is empty.



Coordinate system

$p \rightarrow$ momentum of the particle

 $\eta \rightarrow$ pseudorapidity of the particle

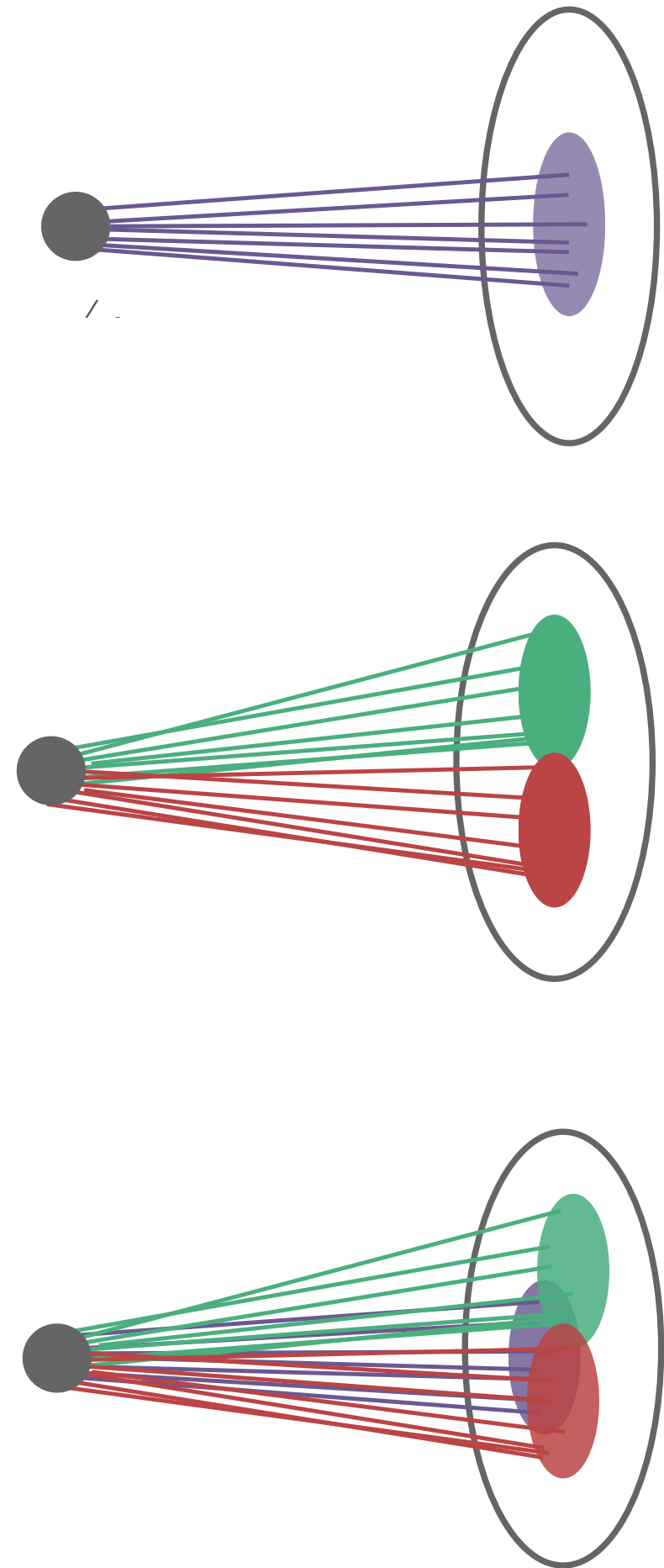
$$\eta = -\log(\tan(\frac{\theta}{2}))$$

 $\phi \rightarrow$ azimuthal angle

$\Delta \rightarrow$ distance from the jet axis

Dataset & Dimensionality Reduction

Jet = spray of particles



300 features

100

Diagram illustrating a 3D grid structure, likely representing a phase space volume element. The grid is composed of small cubes. The top face is labeled with p_T , $\Delta\eta$, and $\Delta\phi$. A vertical double-headed arrow on the left indicates the extent in the third dimension.

Training data = known physics

HEP data

Standard Model

beyond Standard Model

Test on theoretical assumptions of anomalies

encoder

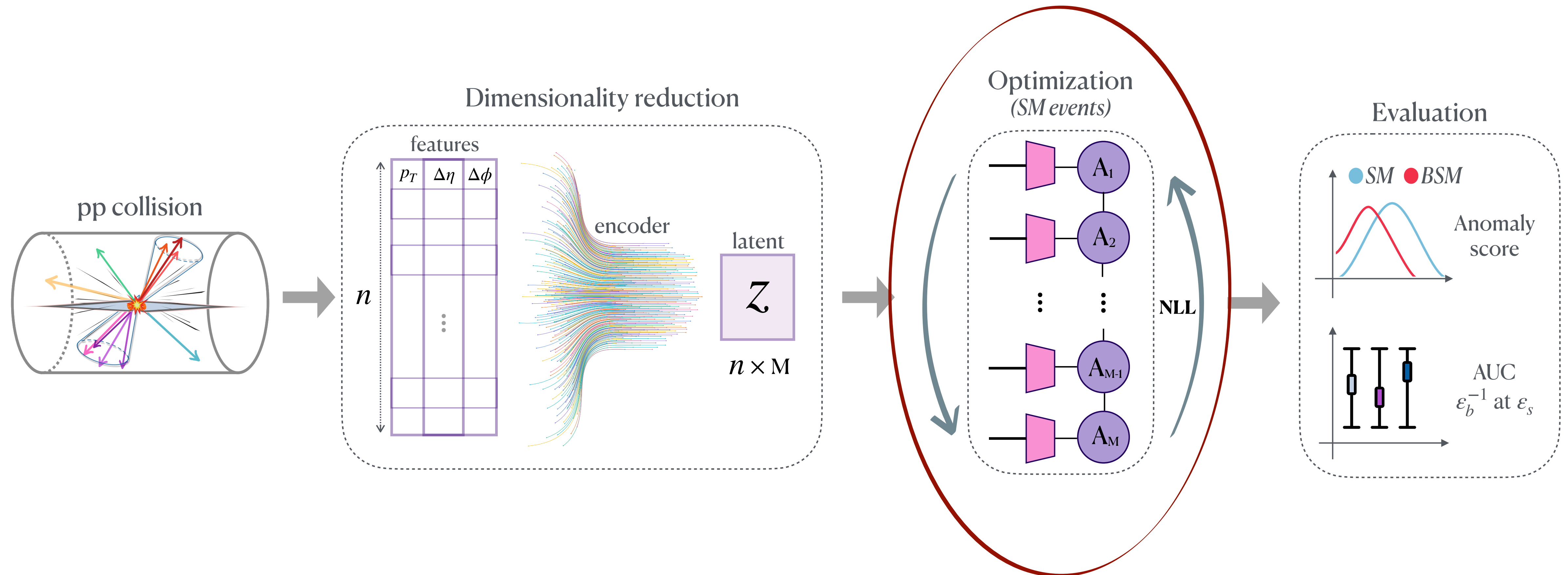
latent

Z

$$n \times M$$

Dataset info - [zenodo7673769](#)

TNs for Anomaly Detection in the Latent Space of Proton Collision Events at the LHC



Discrete Probabilistic Anomaly Detection

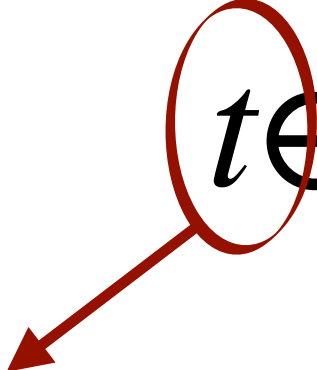
Discrete Probabilistic Anomaly Detection

Data distribution

$$|\Psi\rangle = \sum_{t \in \tau} \Psi(t) |t\rangle$$

Discrete Probabilistic Anomaly Detection

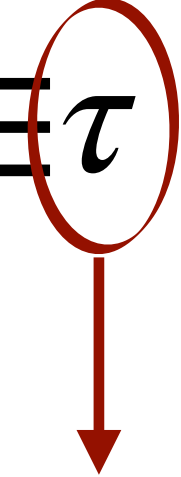
Data distribution

$$|\Psi\rangle = \sum_{t \in \tau} \Psi(t) |t\rangle$$


Data point $\rightarrow N$ random variables

Discrete Probabilistic Anomaly Detection

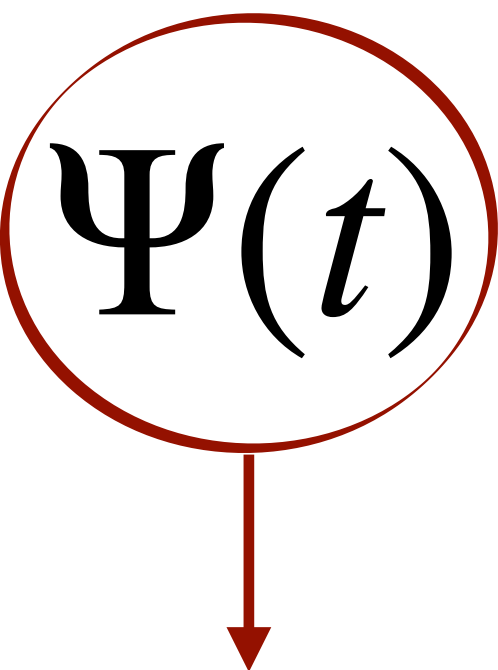
Data distribution

$$|\Psi\rangle = \sum_{t \in \tau} \Psi(t) |t\rangle$$


All possible combinations of N variables

Discrete Probabilistic Anomaly Detection

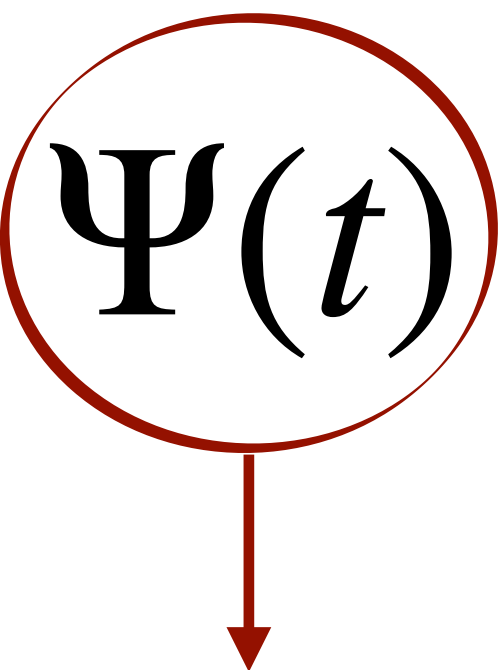
Data distribution

$$|\Psi\rangle = \sum_{t \in \tau} \Psi(t) |t\rangle$$


Amplitude associated with each
unique data point configuration

Discrete Probabilistic Anomaly Detection

Data distribution

$$|\Psi\rangle = \sum_{t \in \tau} \Psi(t) |t\rangle$$


Coefficient ~~Amplitude~~ associated with each
unique data point configuration

Discrete Probabilistic Anomaly Detection

Data distribution

$$|\Psi\rangle = \sum_{t \in \tau} \Psi(t) |t\rangle$$

↓
basis state

Discrete Probabilistic Anomaly Detection

$$|\Psi\rangle = \sum_{t \in \tau} \Psi(t) |t\rangle$$

$$t \equiv (t_1, t_2, \dots, t_N)$$

Possible data point

Discrete Probabilistic Anomaly Detection

$$|\Psi\rangle = \sum_{t \in \tau} \Psi(t) |t\rangle$$

Born rule - quantum mechanics

$$t \equiv (t_1, t_2, \dots, t_N) \longrightarrow P(t) = \frac{1}{Z} |\Psi(t)|^2$$

Possible data point

Probability of that data point happening

Discrete Probabilistic Anomaly Detection

$$|\Psi\rangle = \sum_{t \in \tau} \Psi(t) |t\rangle$$

$$t \equiv (t_1, t_2, \dots, t_N) \longrightarrow P(t) = \frac{1}{Z} |\Psi(t)|^2$$

normalization factor

$$Z = \sum_{t \in \tau} |\Psi(t)|^2$$

Discrete Probabilistic Anomaly Detection

$$|\Psi\rangle = \sum_{t \in \tau} \Psi(t) |t\rangle$$

Represent as MPS

$$t \equiv (t_1, t_2, \dots, t_N) \longrightarrow P(t) = \frac{1}{Z} |\Psi(t)|^2$$

normalization factor

$$Z = \sum_{t \in \tau} |\Psi(t)|^2$$

Discrete Probabilistic Anomaly Detection

$$|\Psi\rangle = \sum_{t \in \tau} \Psi(t) |t\rangle$$

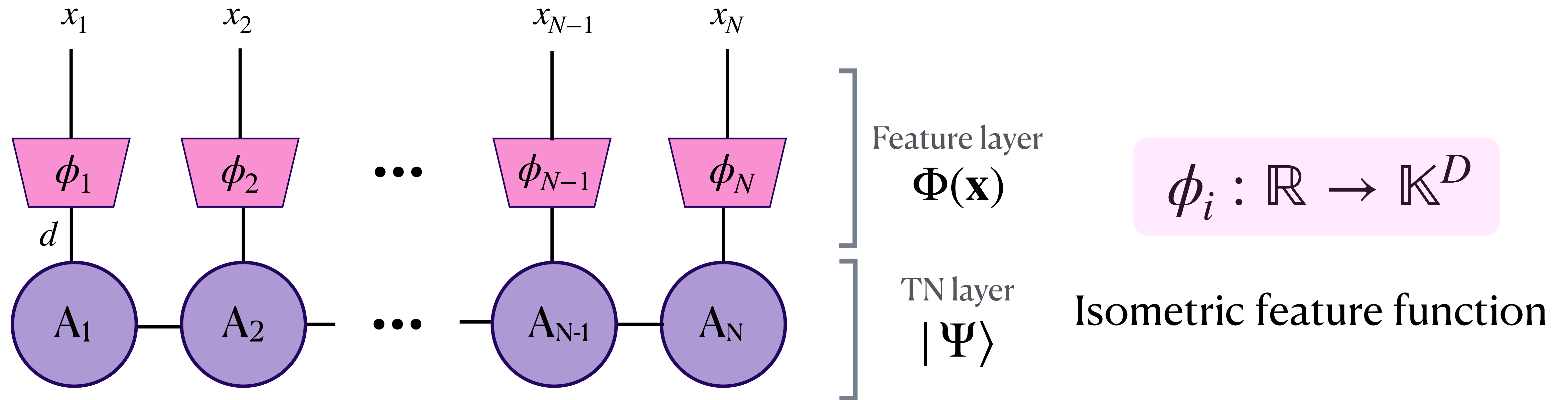
$$P(t) = \frac{1}{Z} |\Psi(t)|^2$$

$$Z = \sum_{t \in \tau} |\Psi(t)|^2$$

Matrix Product State

$$\Psi(t_1, t_2, \dots, t_N) = A^{(1)t_1} A^{(2)t_2} \dots A^{(N)t_N}$$

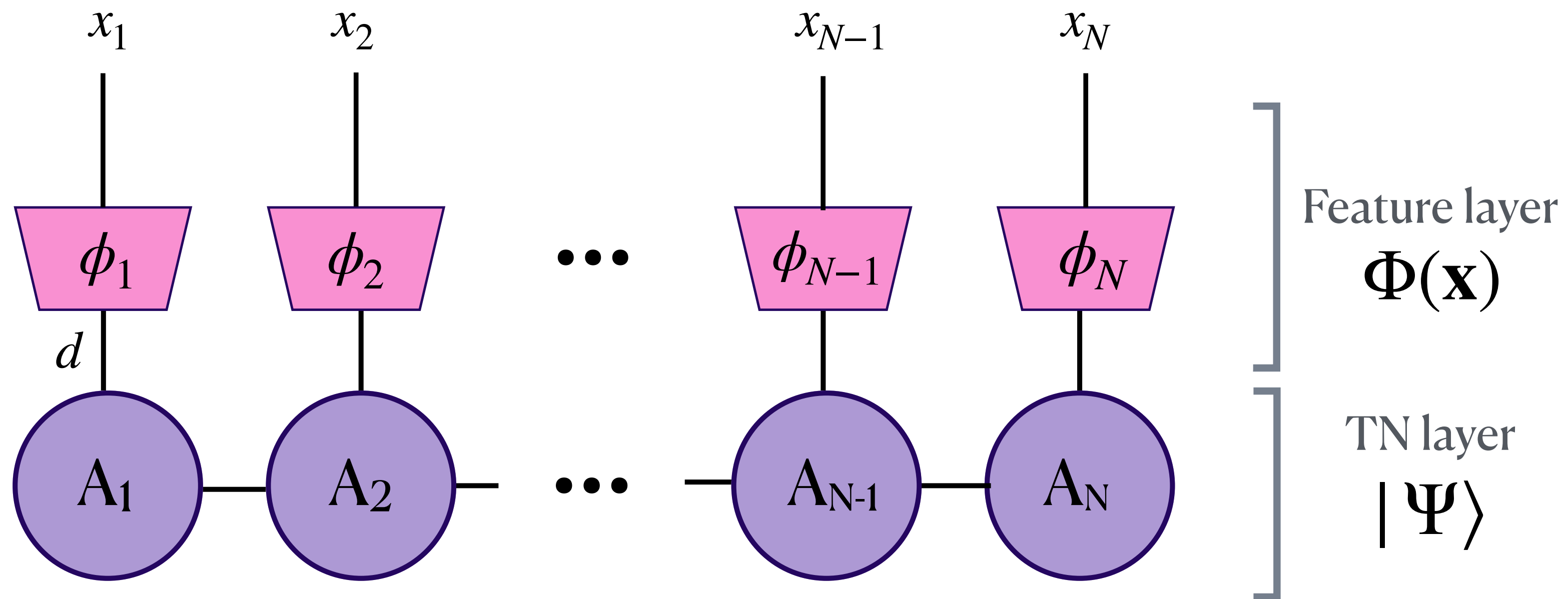
Continuous-valued MPS for Anomaly Detection



Latent feature space - continuous

- need to translate it to a discrete space with feature layer

Continuous-valued MPS for Anomaly Detection

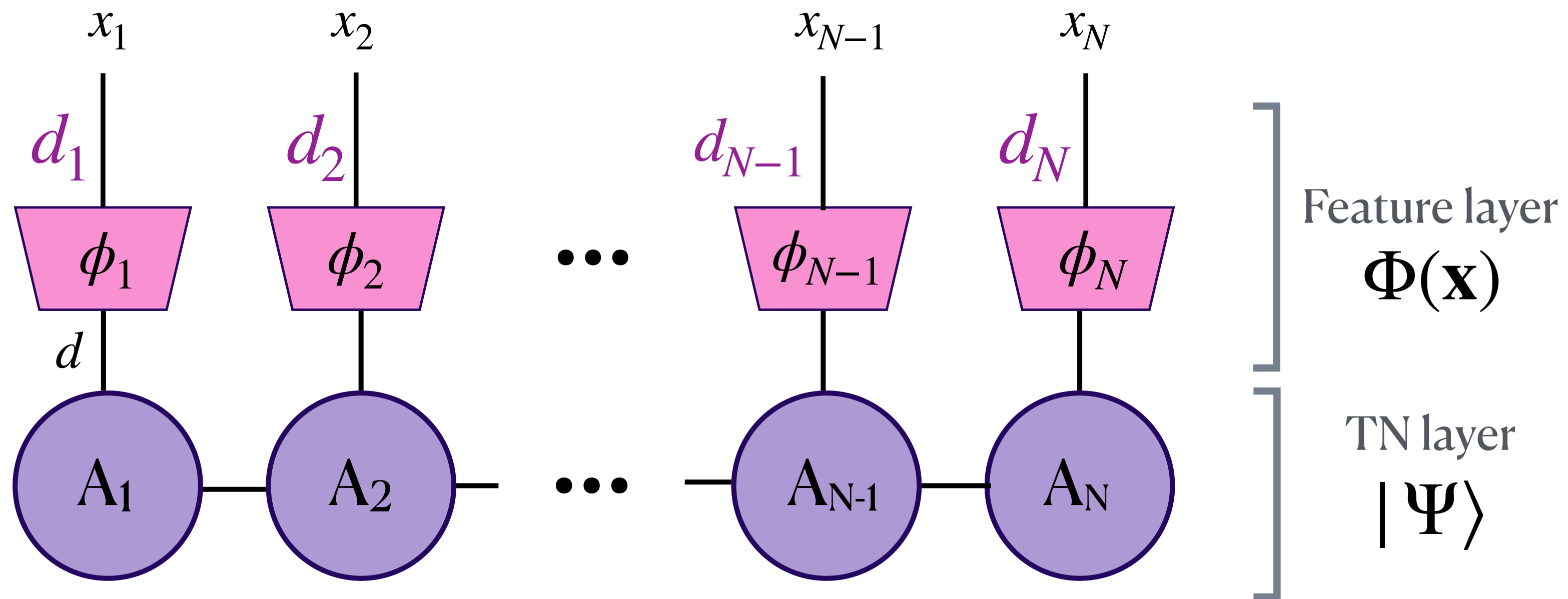


$$\Phi(\mathbf{x}) = \bigotimes_{i=1}^N \phi_i(x_i)$$

Latent feature space - continuous

- need to translate it to a discrete space with feature layer

Continuous-valued MPS for Anomaly Detection

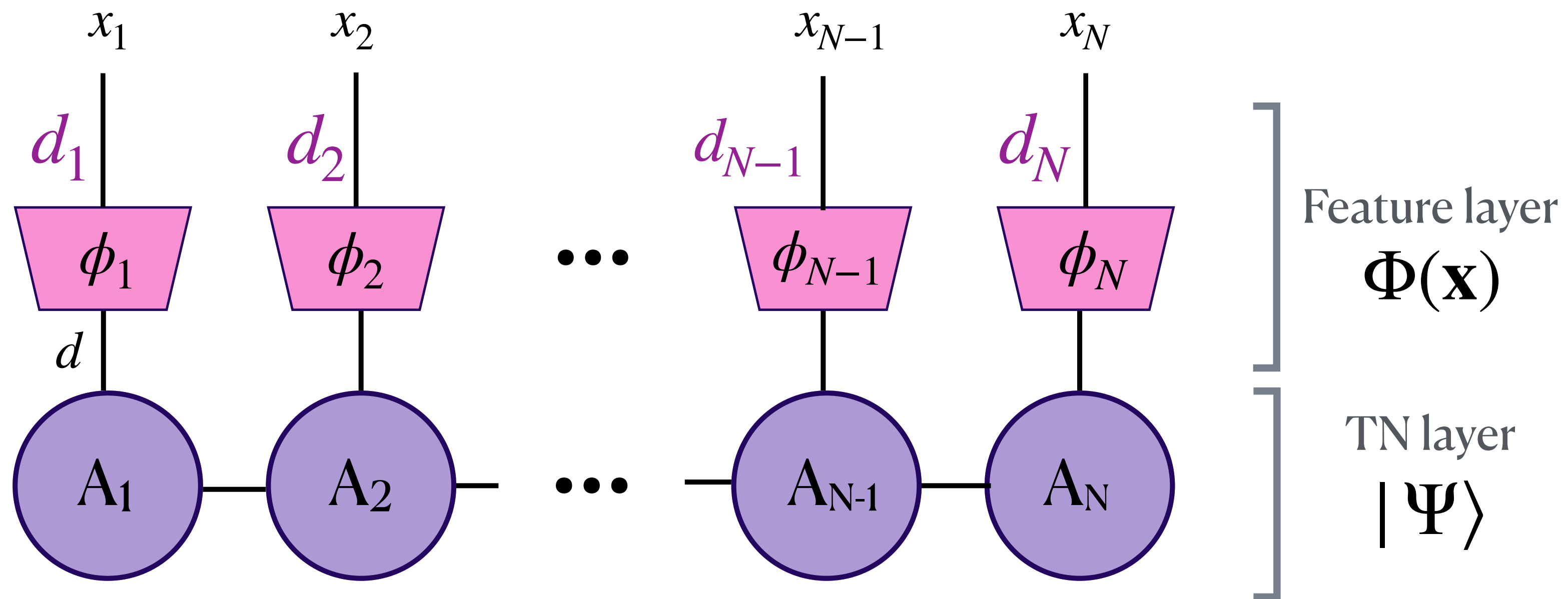


$$\Phi(\mathbf{x}) = \bigotimes_{i=1}^N \phi_i(x_i)$$

Latent feature space - continuous

- need to translate it to a discrete space with feature layer

Continuous-valued MPS for Anomaly Detection



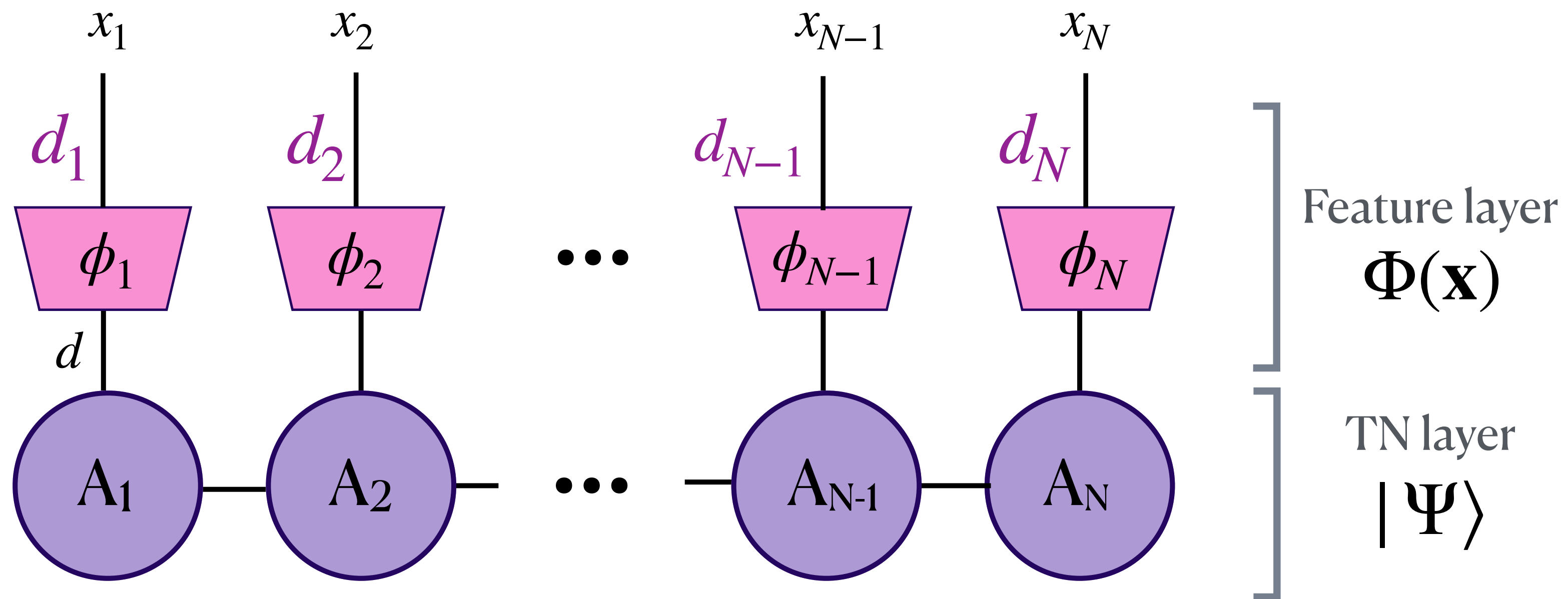
amplitude

$$\theta(\mathbf{x}) = \langle \Phi(\mathbf{x}) | \Psi \rangle$$

Latent feature space - continuous

- need to translate it to a discrete space with feature layer

Continuous-valued MPS for Anomaly Detection



Outcome Probability

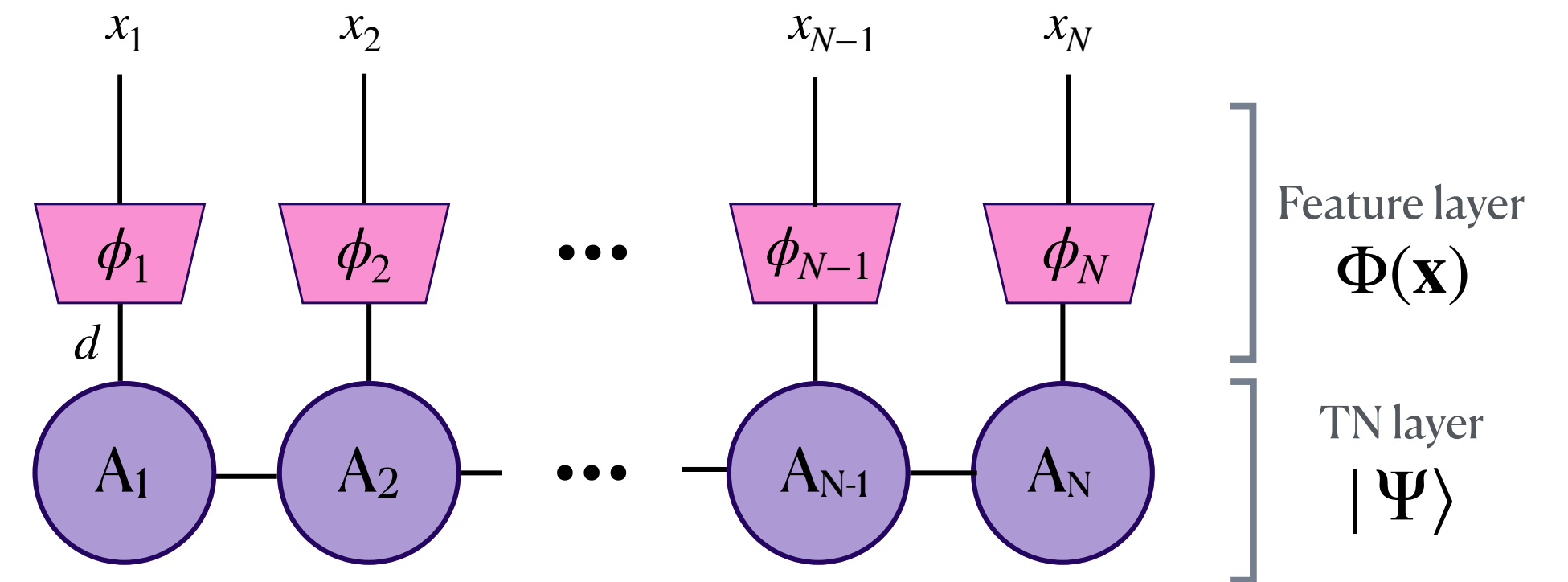
$$P(\mathbf{x}) = |\langle \Phi(\mathbf{x}) | \Psi \rangle|^2$$

Latent feature space - continuous

- need to translate it to a discrete space with feature layer

Choice of feature function for feature layer

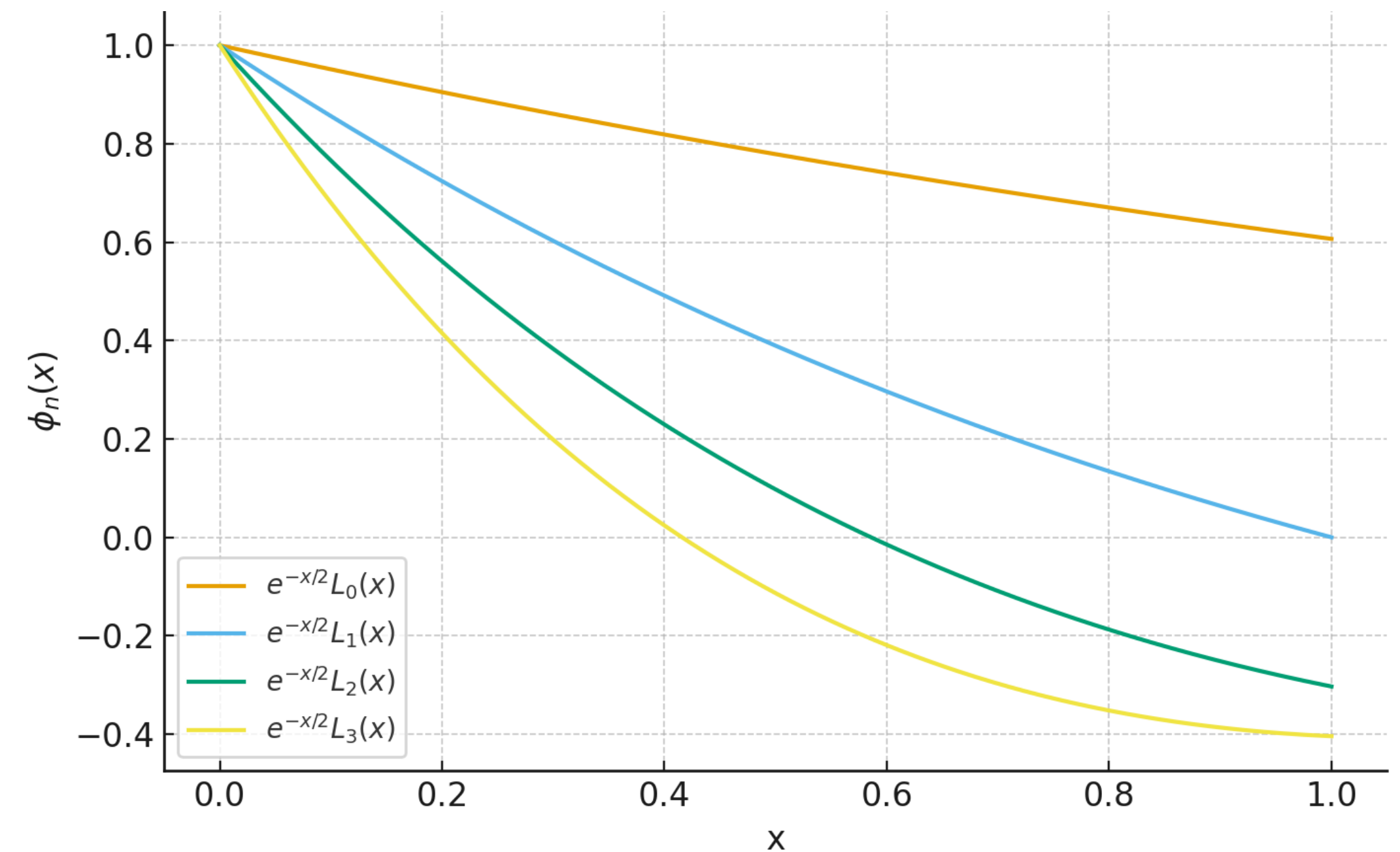
Choice with hyperparameter optimization



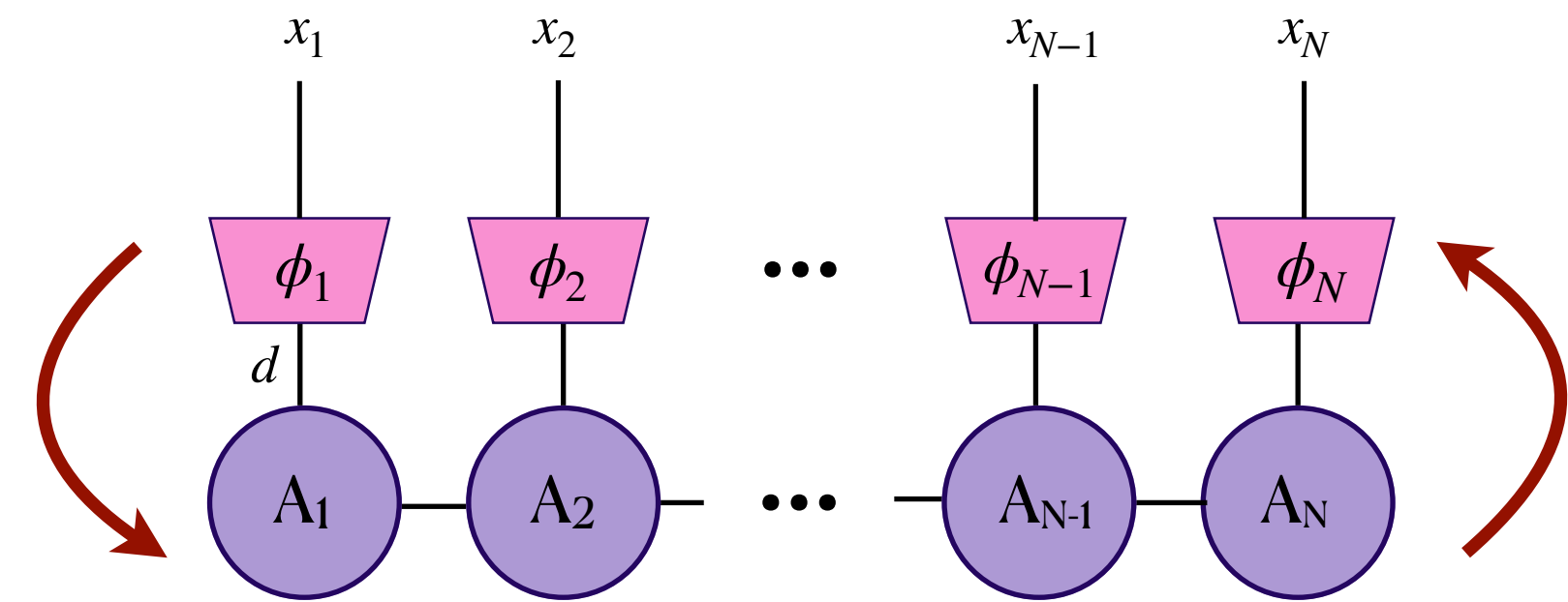
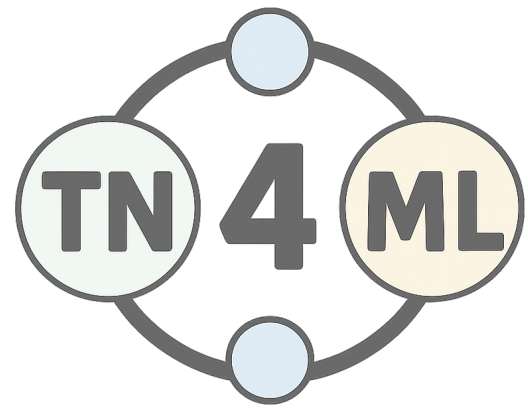
Laguerre Polynomial

$$L_n(x) = \sum_{k=0}^n \frac{(-1)^k}{k!} \binom{n}{k} x^k$$

$$\phi(x) = e^{-x/2} L_n(x)$$

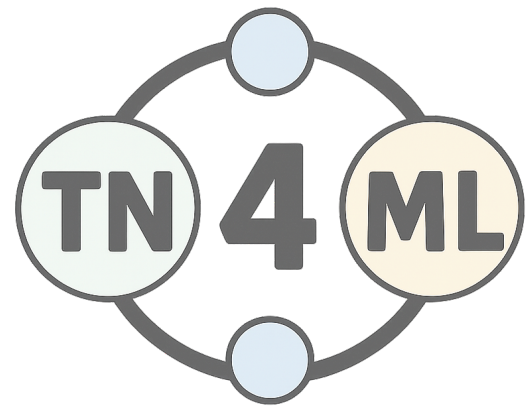


Model Optimization



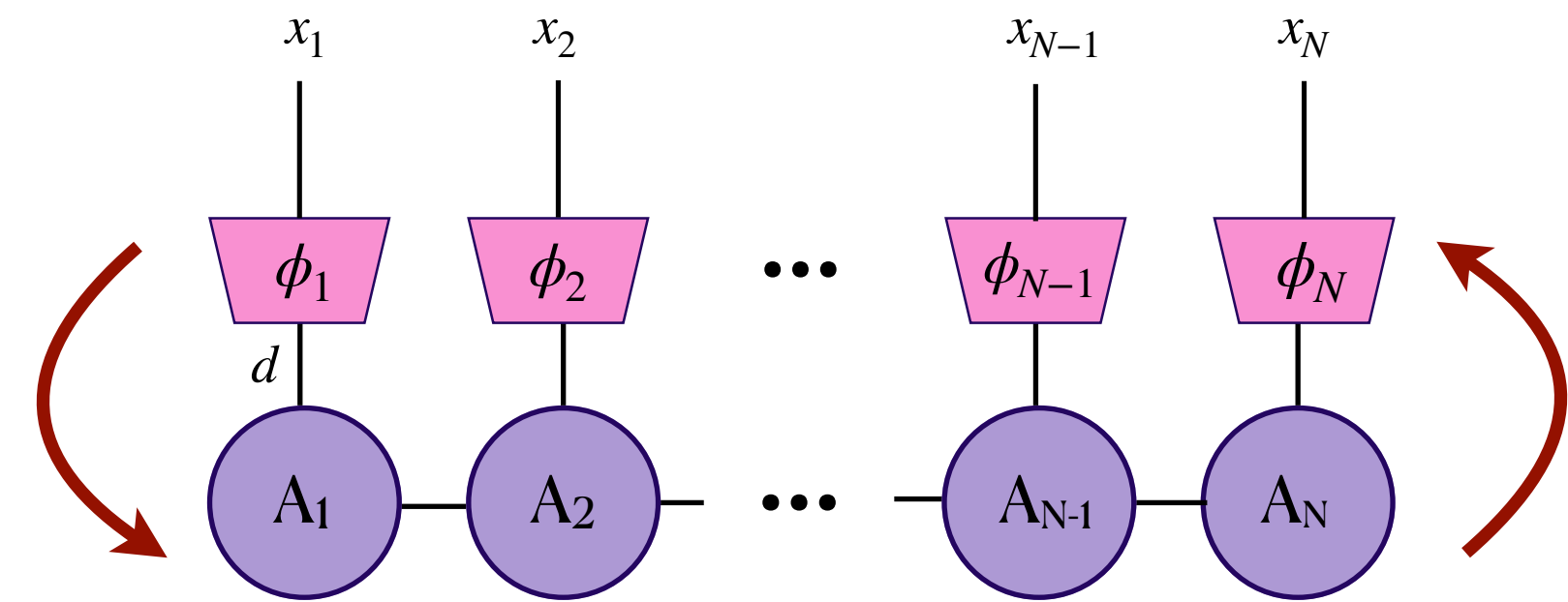
Mini-batch gradient descent

Model Optimization



Negative Log-Likelihood

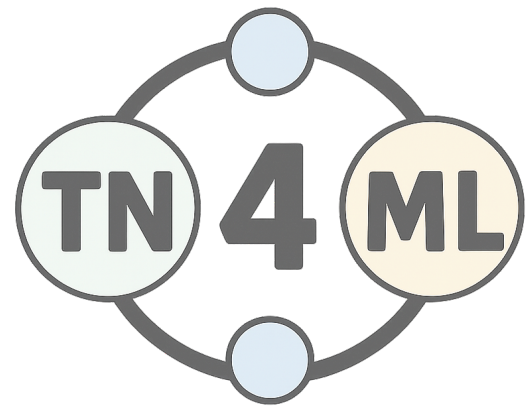
$$L = -\frac{1}{|D|} \sum_{x \in D} \ln P(x)$$



Mini-batch gradient descent

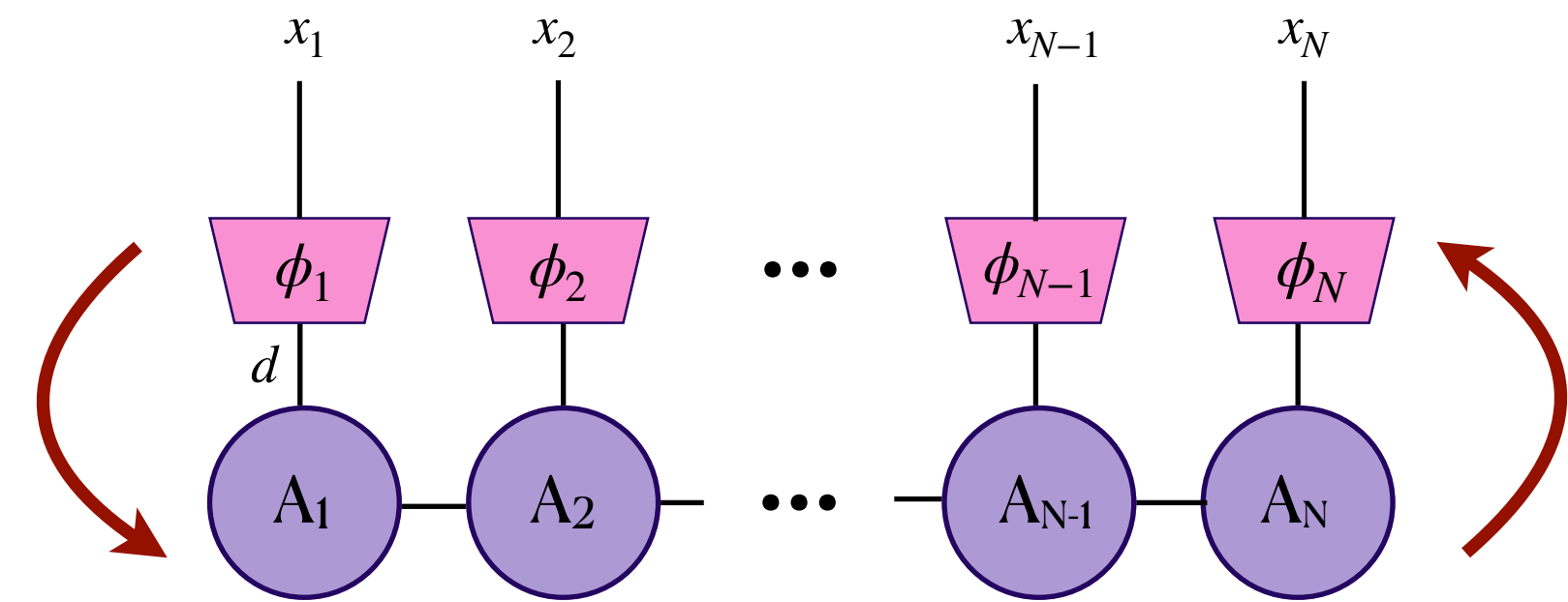
Maximize probability of given input by
minimizing the negative log likelihood

Model Optimization



Negative Log-Likelihood

$$L = -\frac{1}{|D|} \sum_{x \in D} \ln P(x)$$

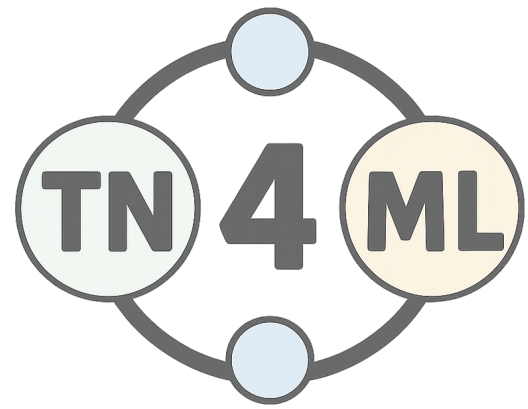


Mini-batch gradient descent

Hyperparameter search

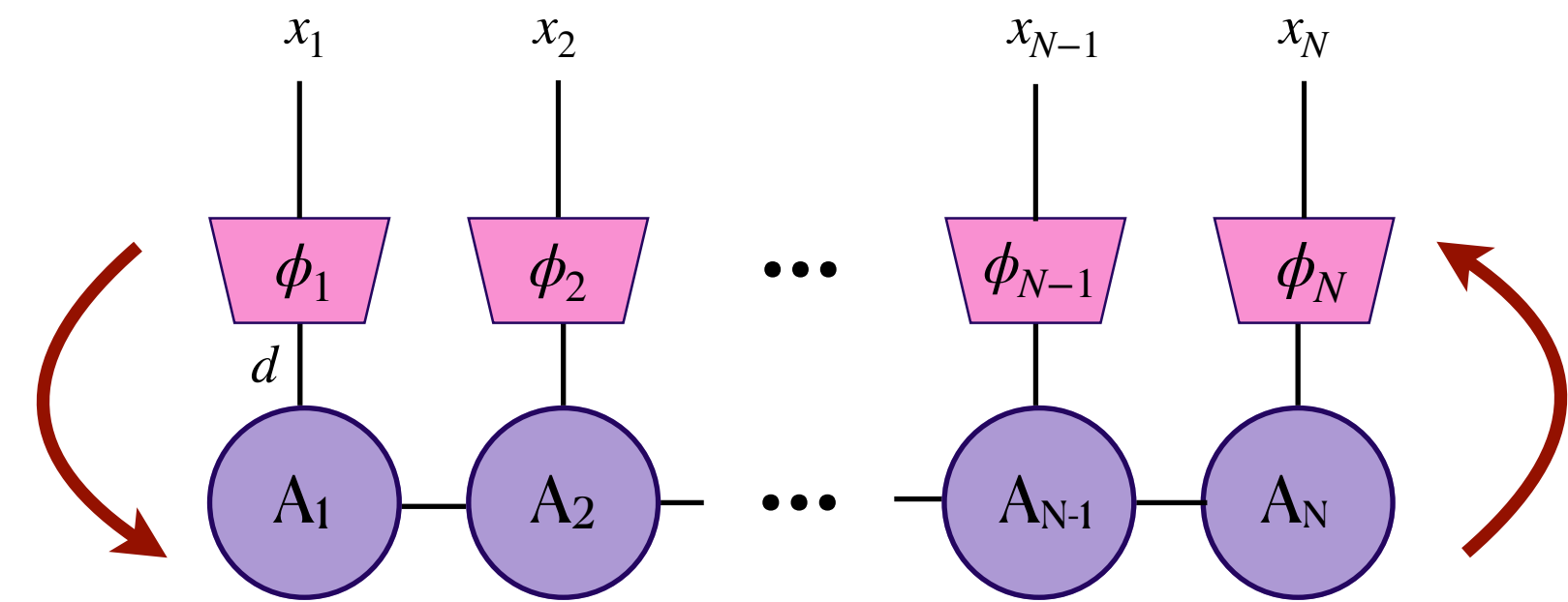
1. **Tensor initializer function**
2. **Feature map function**
3. **Bond dimension**

Model Optimization



Negative Log-Likelihood

$$L = -\frac{1}{|D|} \sum_{x \in D} \ln P(x)$$



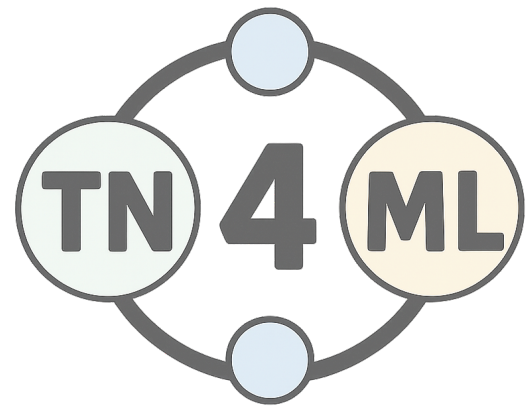
Mini-batch gradient descent

Hyperparameter search

1. **Tensor initializer function**
2. **Feature map function**
3. **Bond dimension**

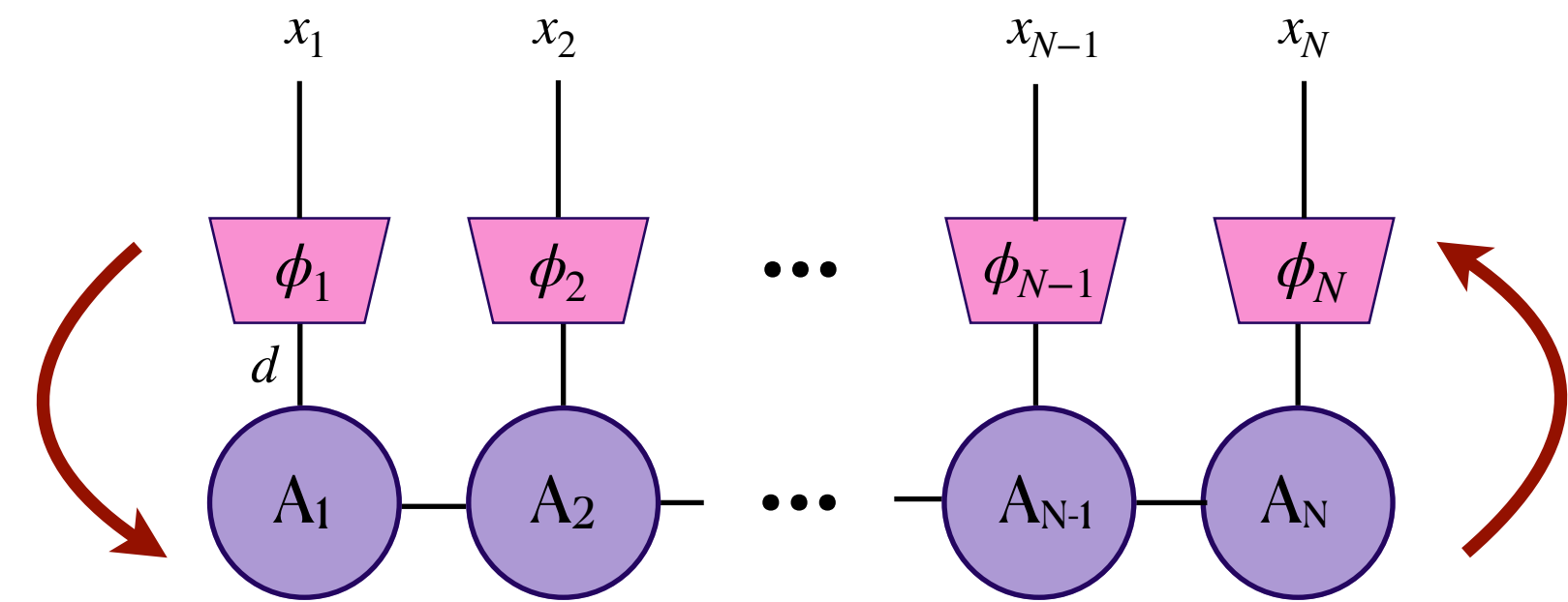
→ unitary initializer

Model Optimization



Negative Log-Likelihood

$$L = -\frac{1}{|D|} \sum_{x \in D} \ln P(x)$$



Mini-batch gradient descent

Hyperparameter search

1. **Tensor initializer function**
2. **Feature map function**
3. **Bond dimension**

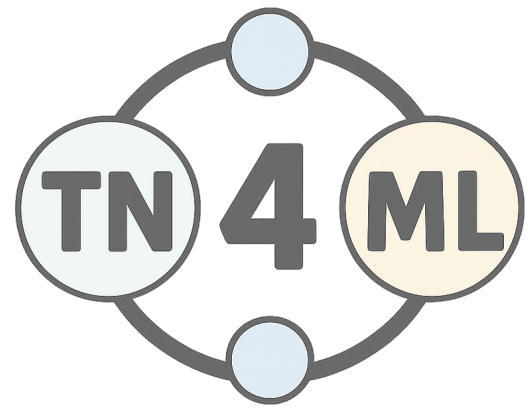


unitary initializer



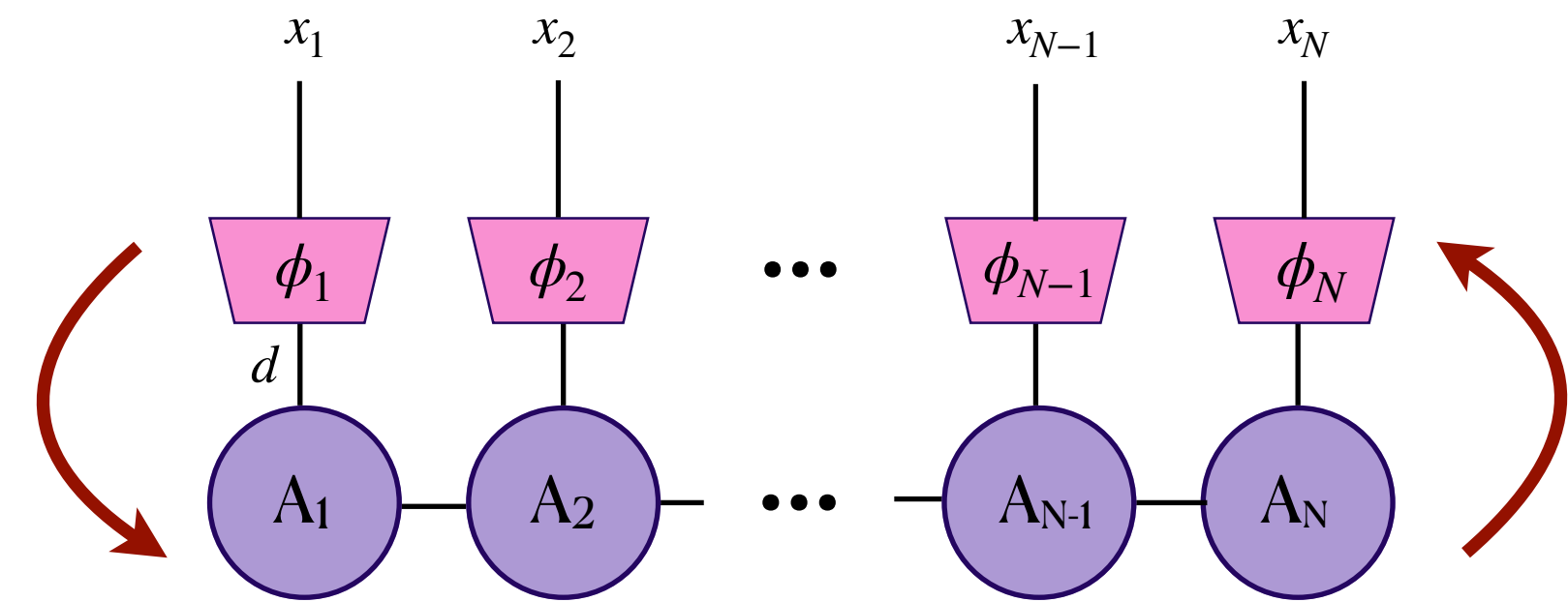
Laguerre polynomial function

Model Optimization



Negative Log-Likelihood

$$L = -\frac{1}{|D|} \sum_{x \in D} \ln P(x)$$



Mini-batch gradient descent

Hyperparameter search

1. **Tensor initializer function**



unitary initializer

2. **Feature map function**



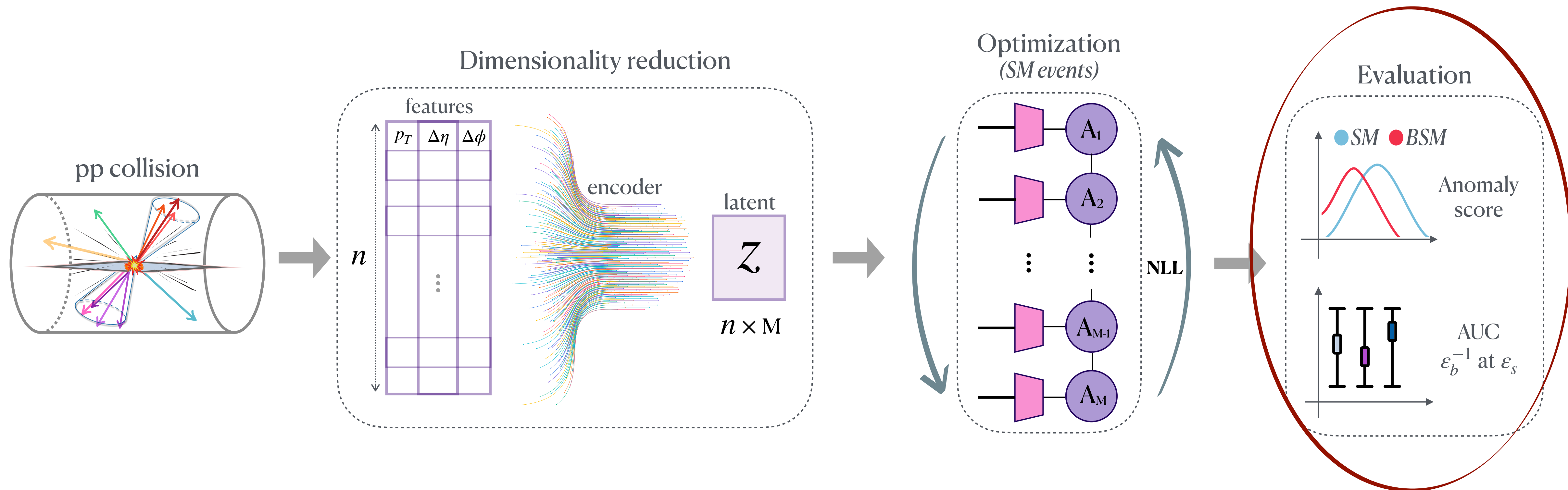
Laguerre polynomial function

3. **Bond dimension**



specific for different latent dimensions

TNs for Anomaly Detection in the Latent Space of Proton Collision Events at the LHC

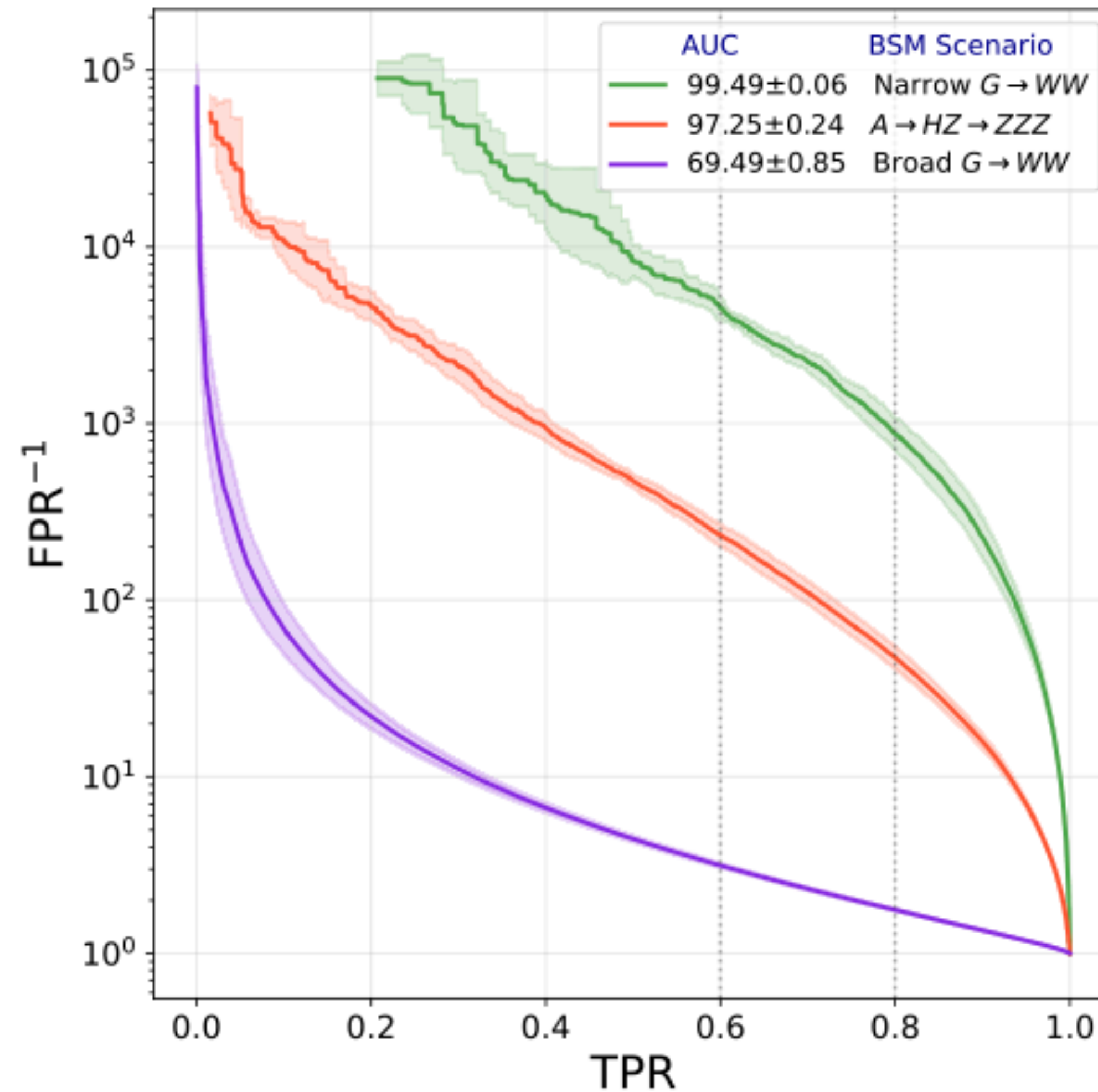


Evaluation (ROC curve)

Beyond SM signatures

- NA $G \rightarrow WW$
- BR $G \rightarrow WW$
- $A \rightarrow HZ \rightarrow ZZZ$

latent = 4, $\chi = 2$



HEP data

Standard
Model

beyond
Standard
Model

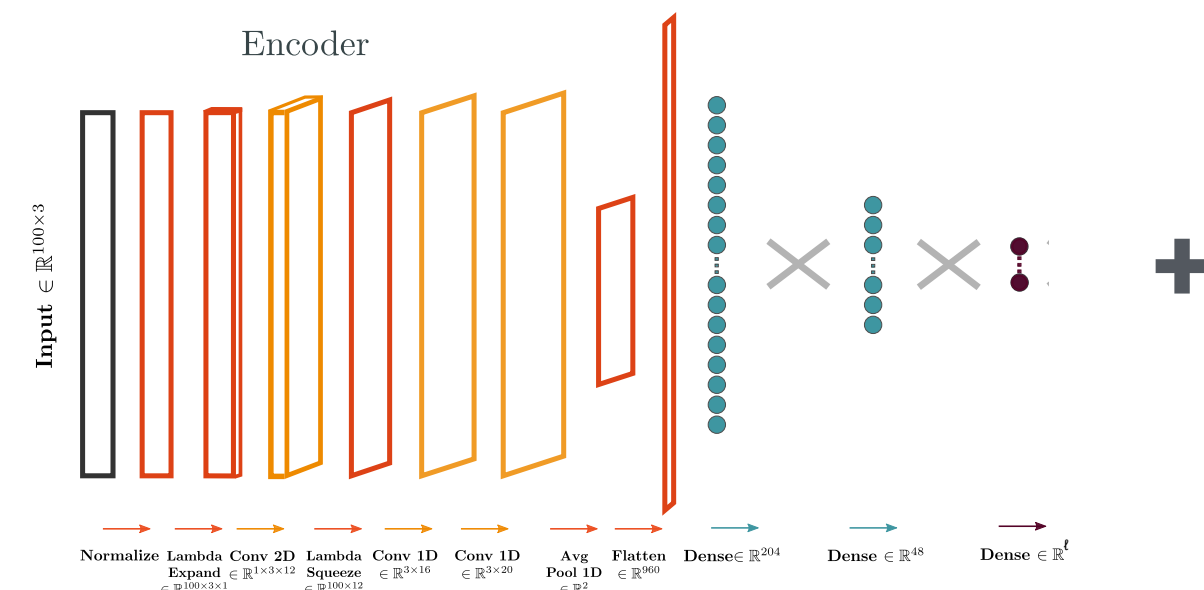
Green anomaly -> easiest one to detect

Purple anomaly -> hardest one to detect

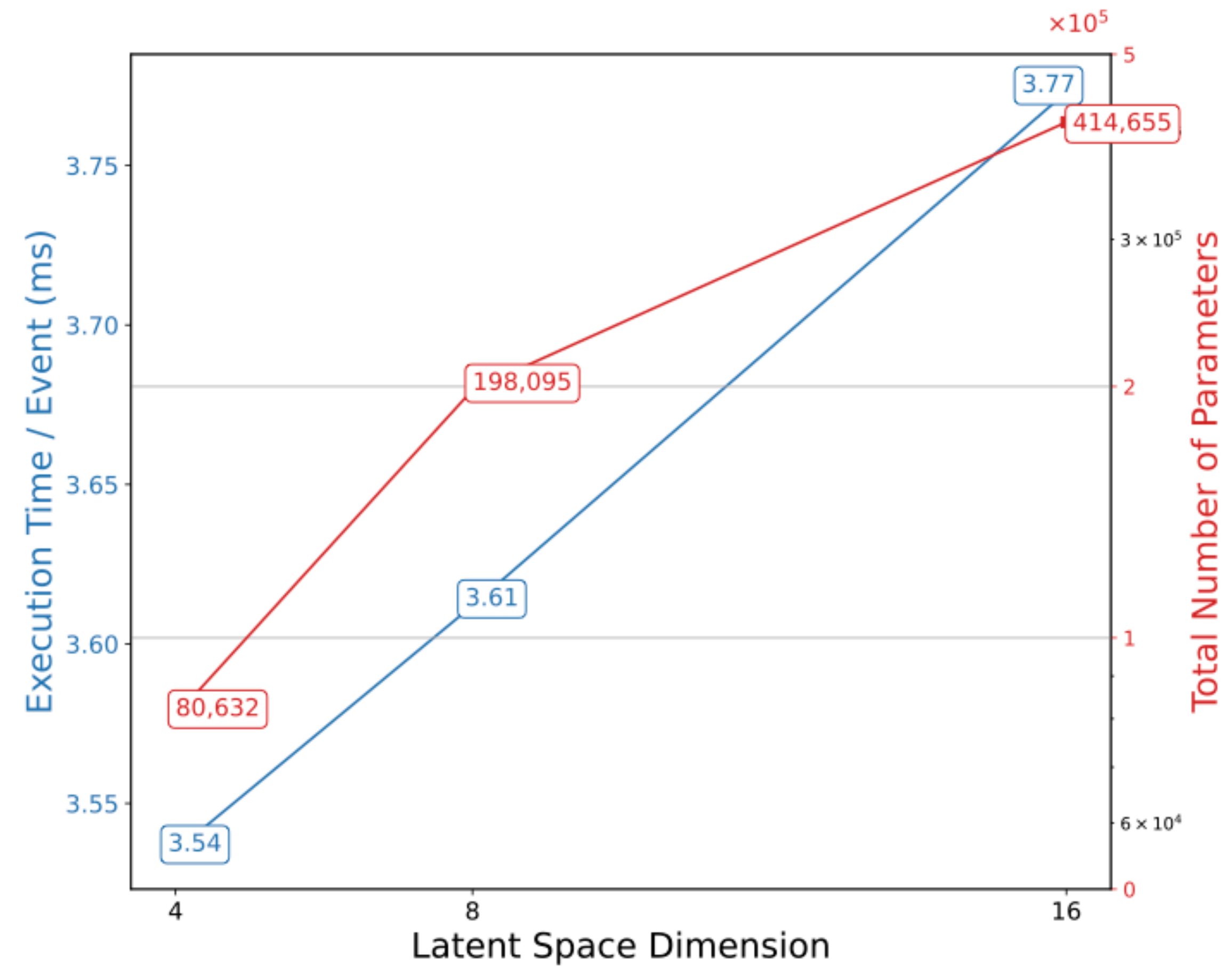
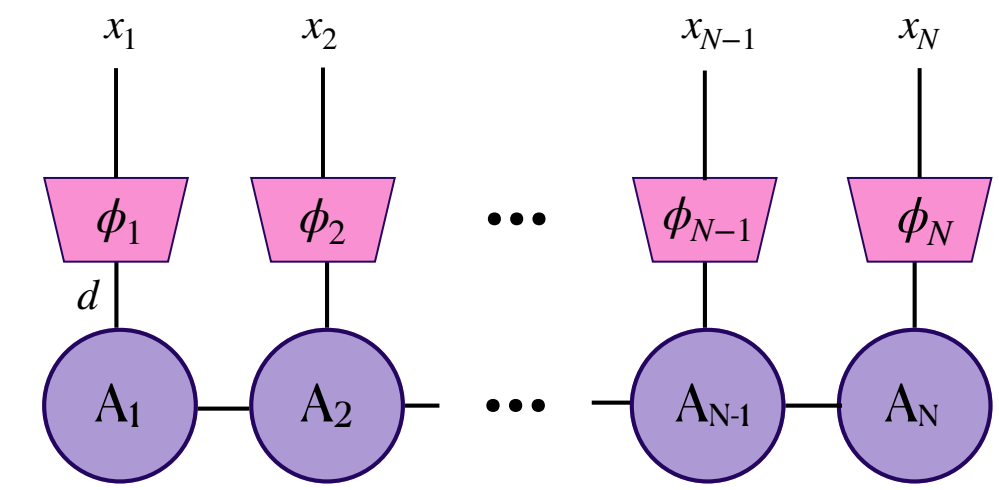
Best result - towards upper right corner

Runtime Evaluation

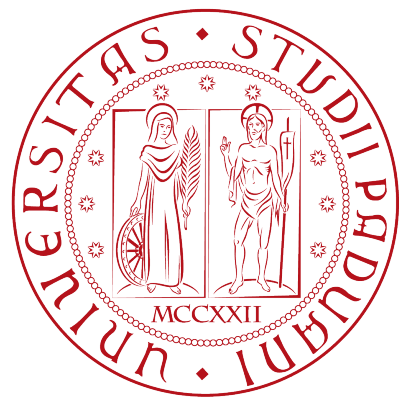
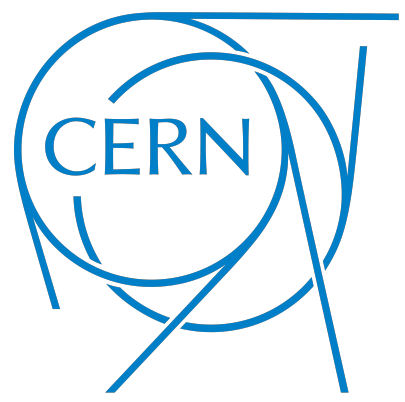
autoencoder



MPS model

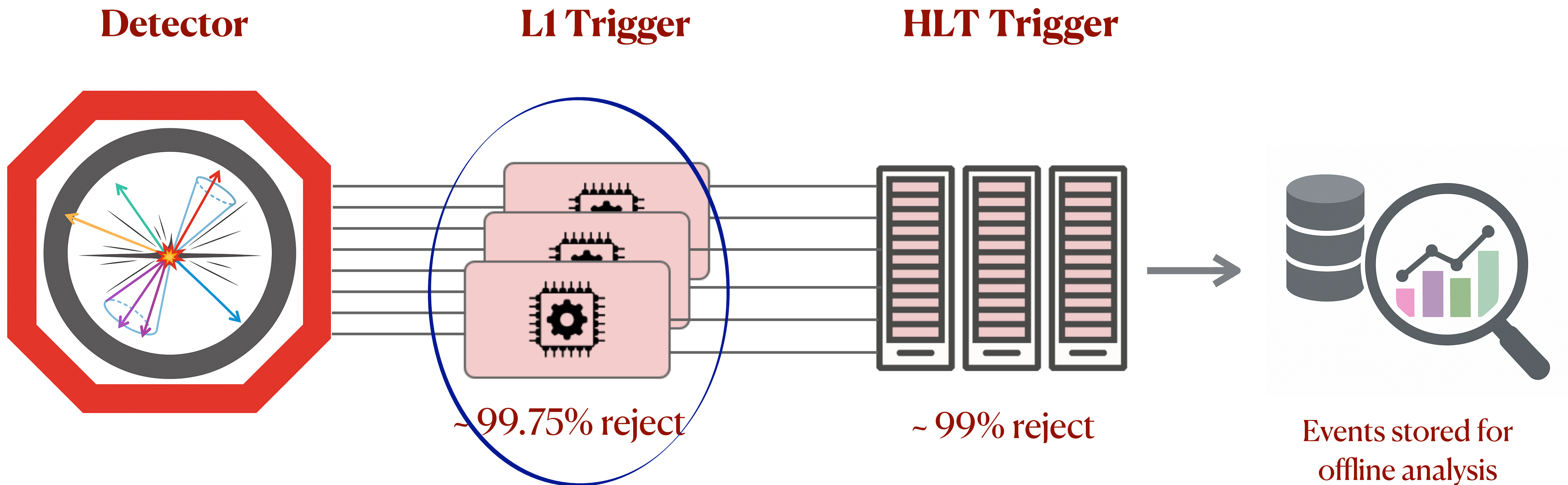


$< \sim 100 \text{ ms} \rightarrow \text{High-Level Trigger}$

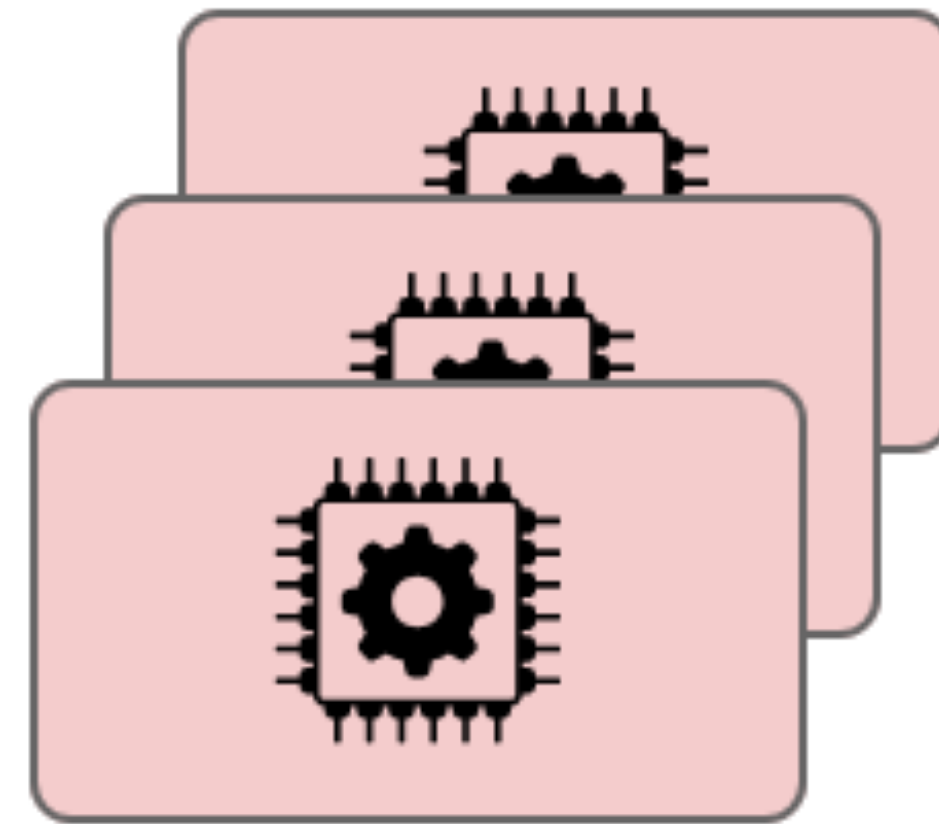


Towards Ultrafast Quantum-Inspired Tensor Network Models for Jet Tagging on FPGAs

Ema Puljak, Alberto Coppi, Lorenzo Borella, Daniel Jaschke, Enrique Rico,
Maurizio Pierini, Jacopo Pazzini, Andrea Triossi, Simone Montangelo



Algorithms needs to run under
 $\mathcal{O}(1) \mu s$



FPGAs

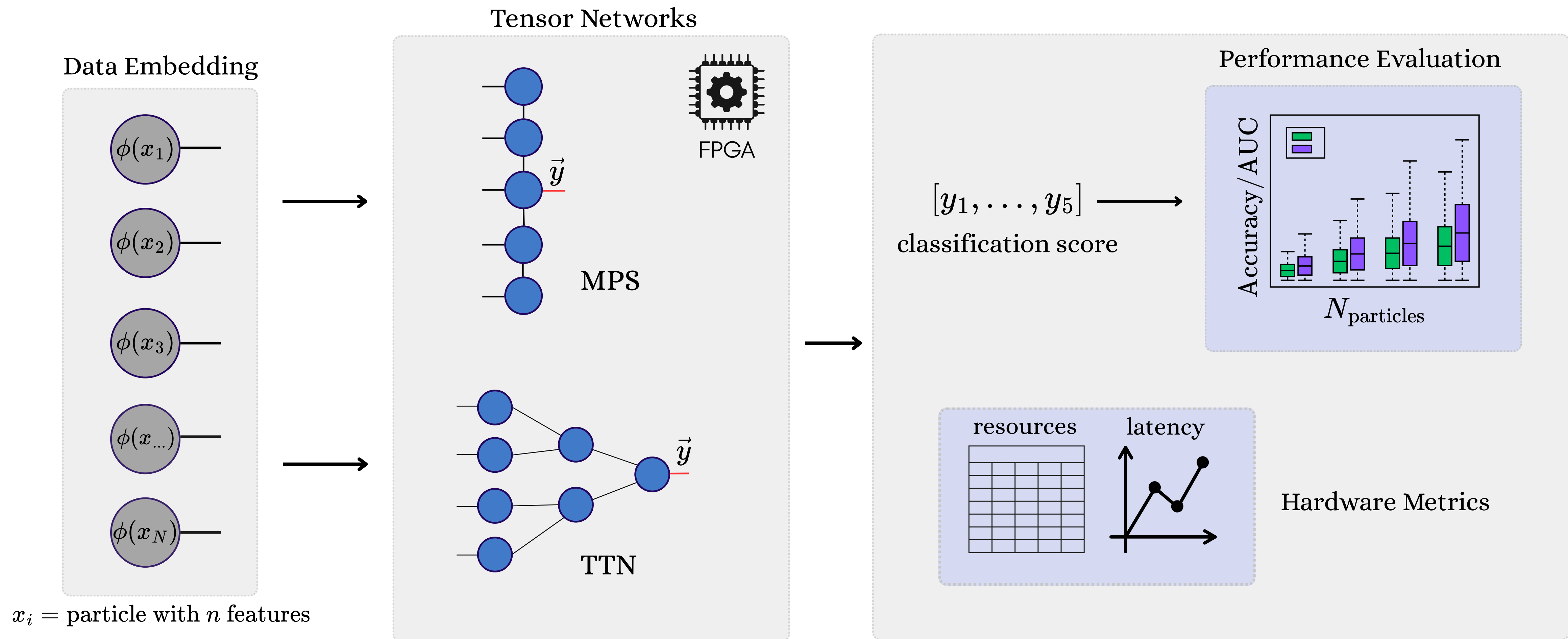
Algorithmic constraints

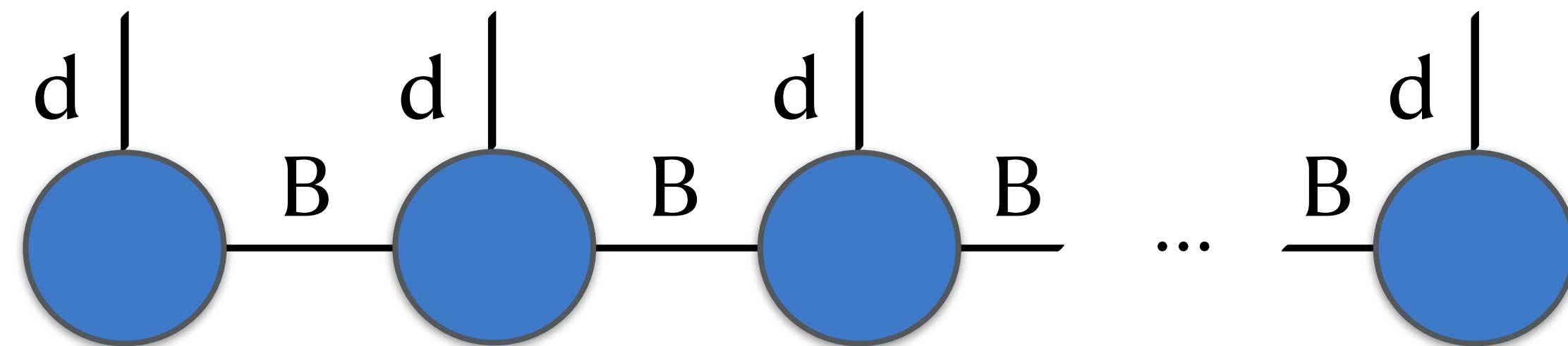
Limited reconstruction of particles

Permutation-robust

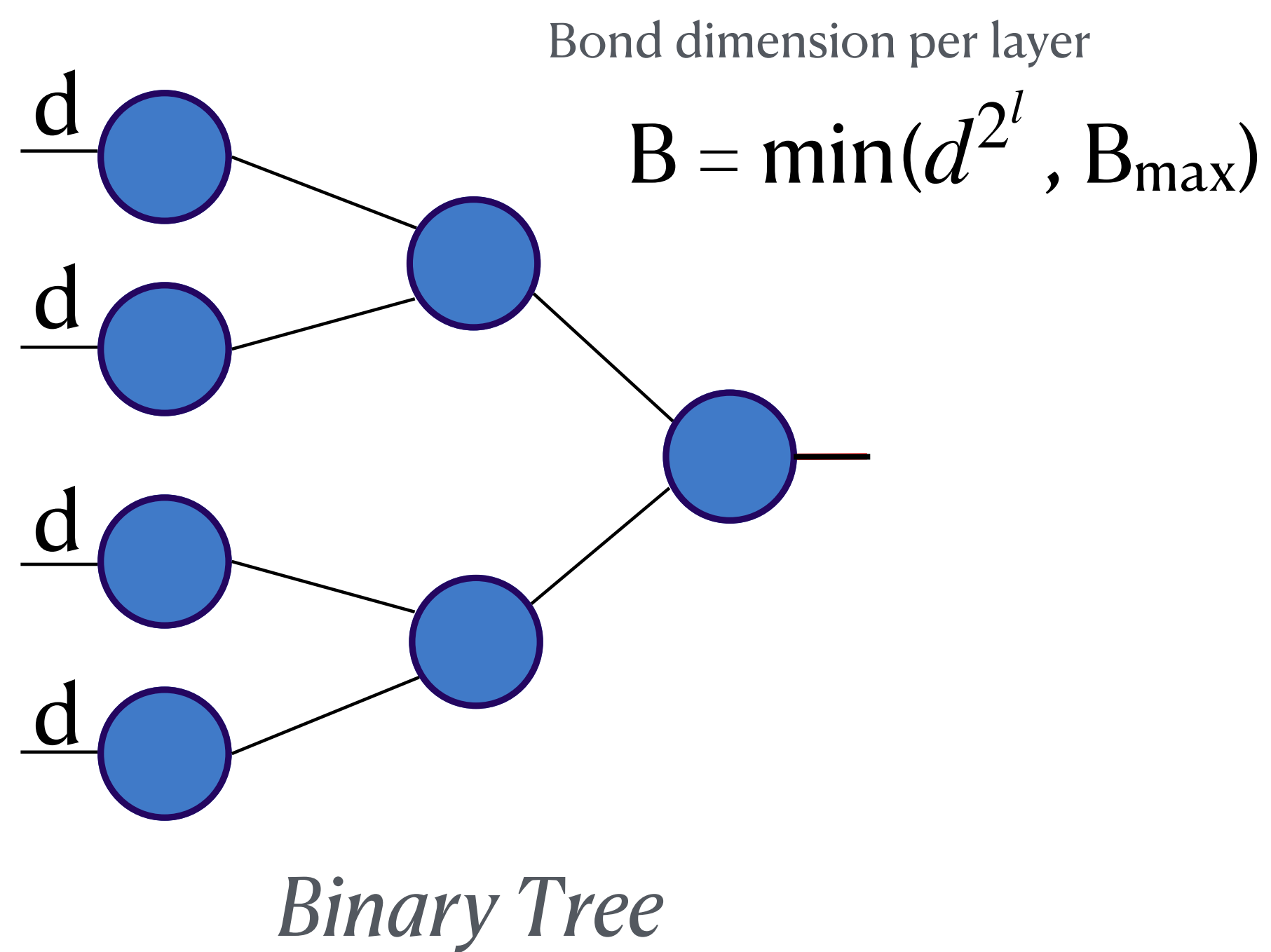
< 100 ns latency

Resource-efficient, compact models





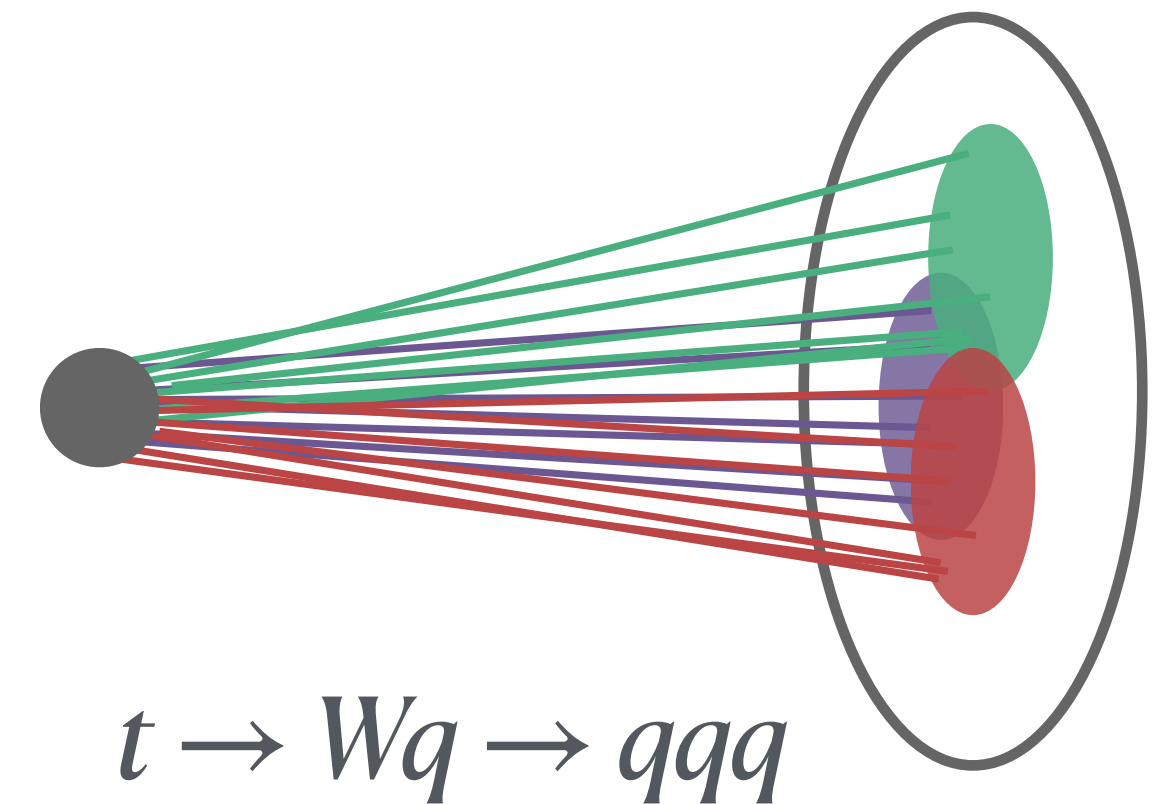
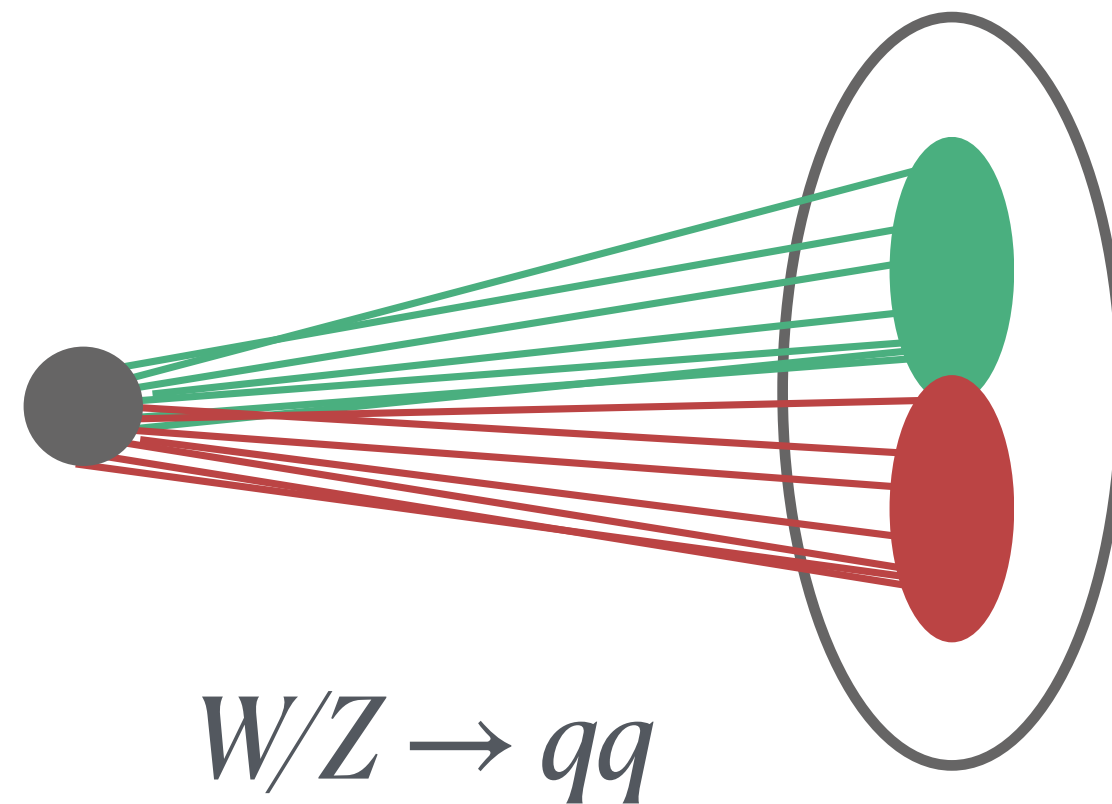
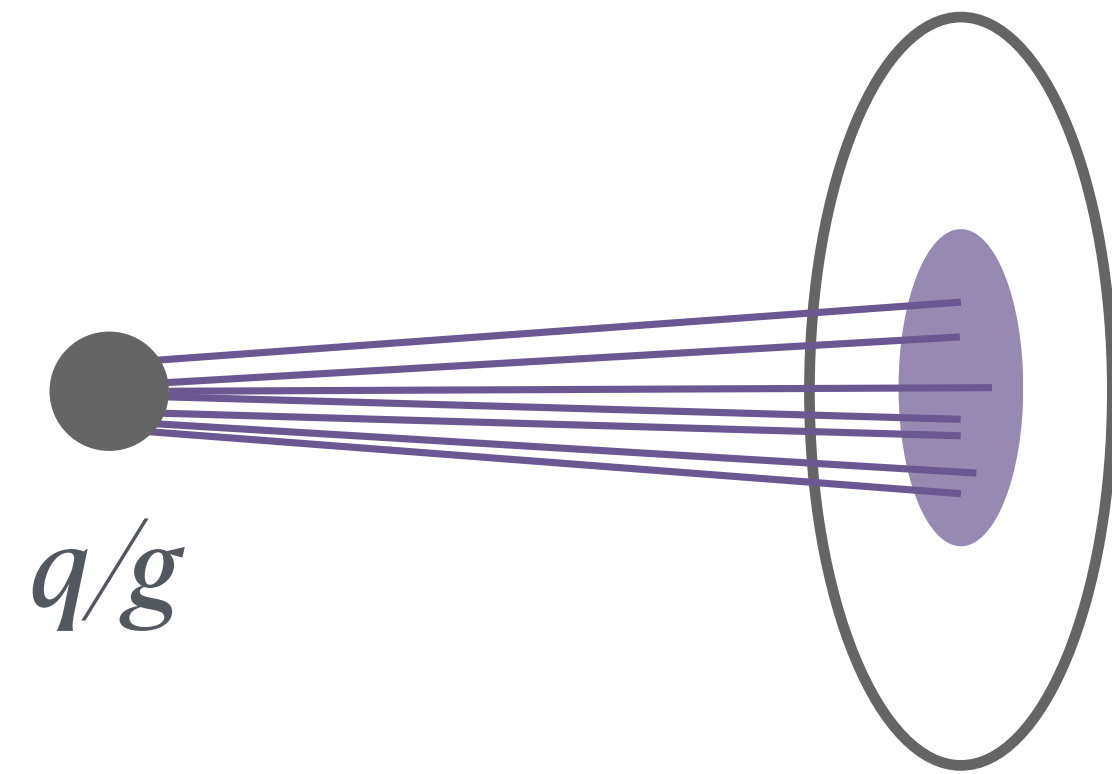
Matrix Product State



Tree Tensor Network

Jet Tagging

5 classes

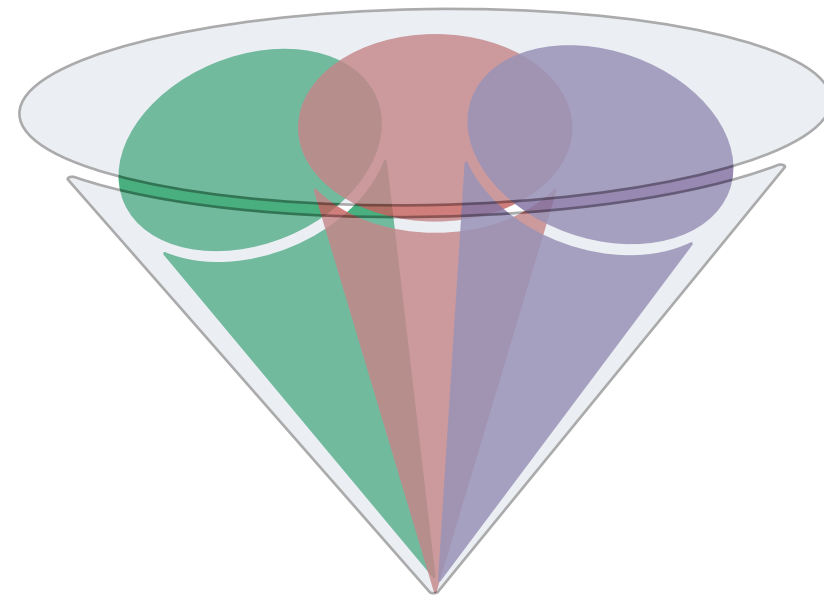


Importance: Critical for trigger decisions and discovery potential

Challenge: Efficient tagging using **low-level** particle information

hls4ml jet dataset

low-level particle constituents



~ 850k jets

150
constituents
per jet

[illegible]

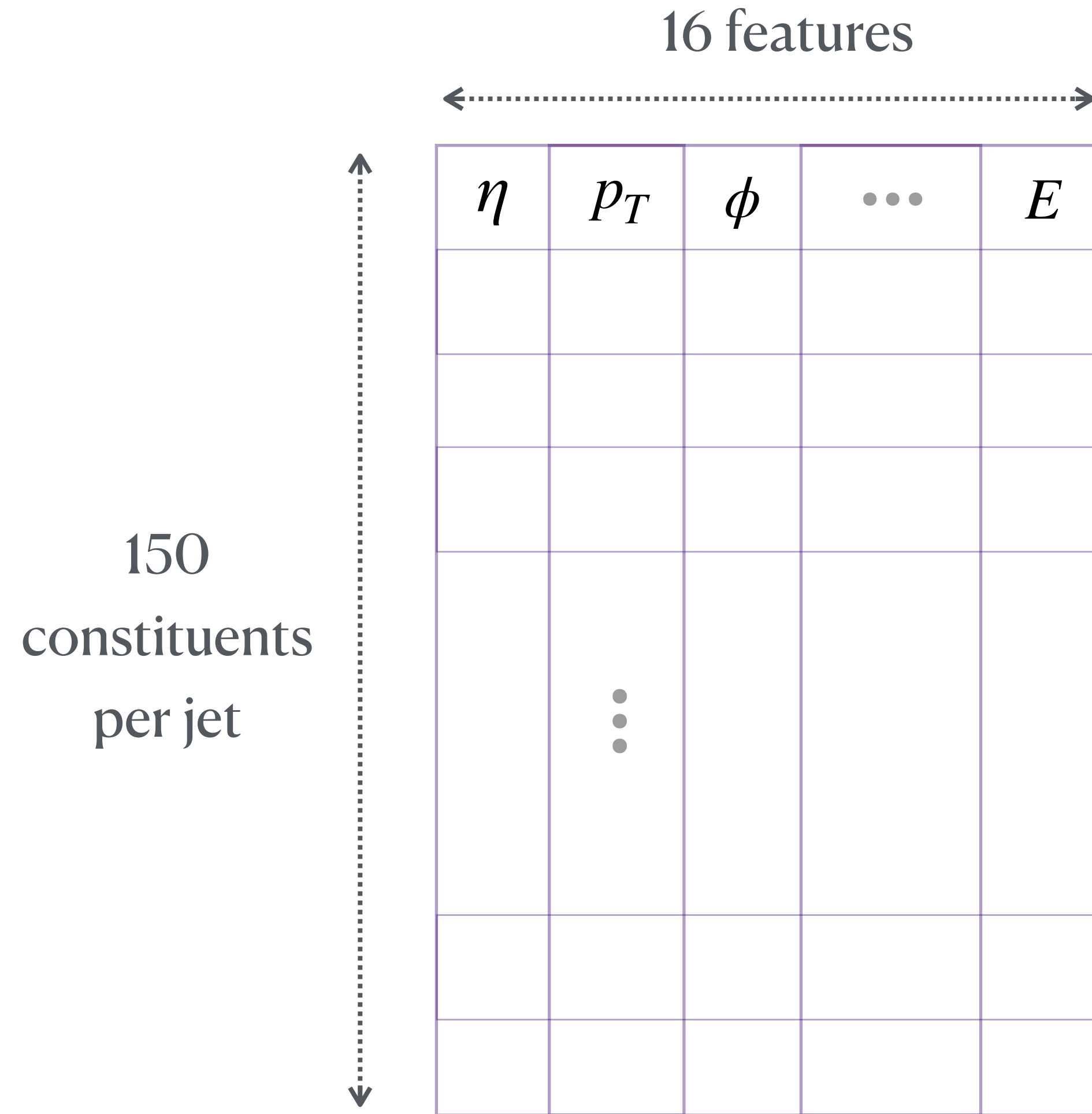
MODEL DESIGN

Data embedding

+

Parametrized TN model

Data Embedding



Data Embedding

(Reduced to most important features)

3 features

p_T	E_{rel}	ΔR
	⋮	

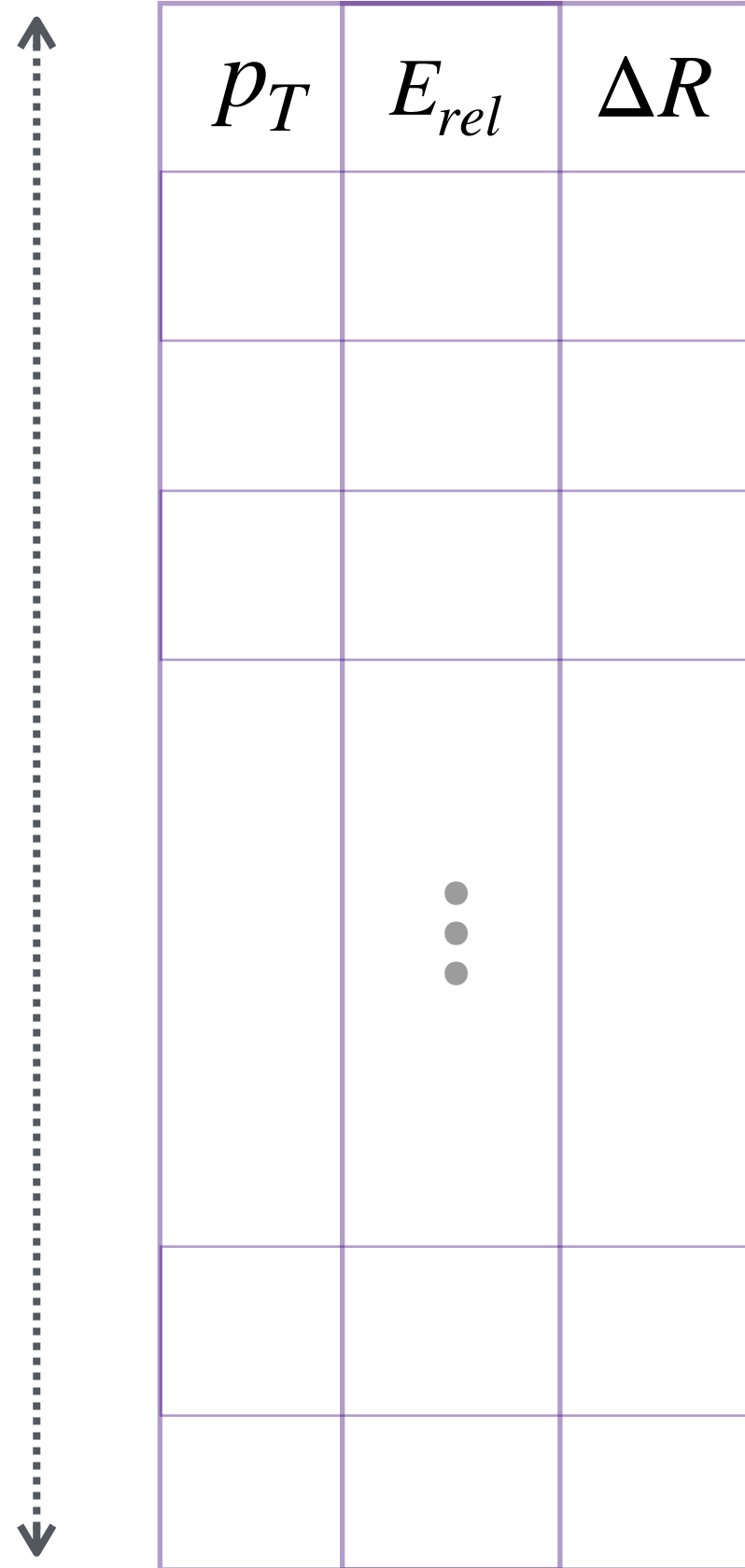
up to 32
constituents
per jet

Data Embedding

(Reduced to most important features)

3 features

up to 32
constituents
per jet

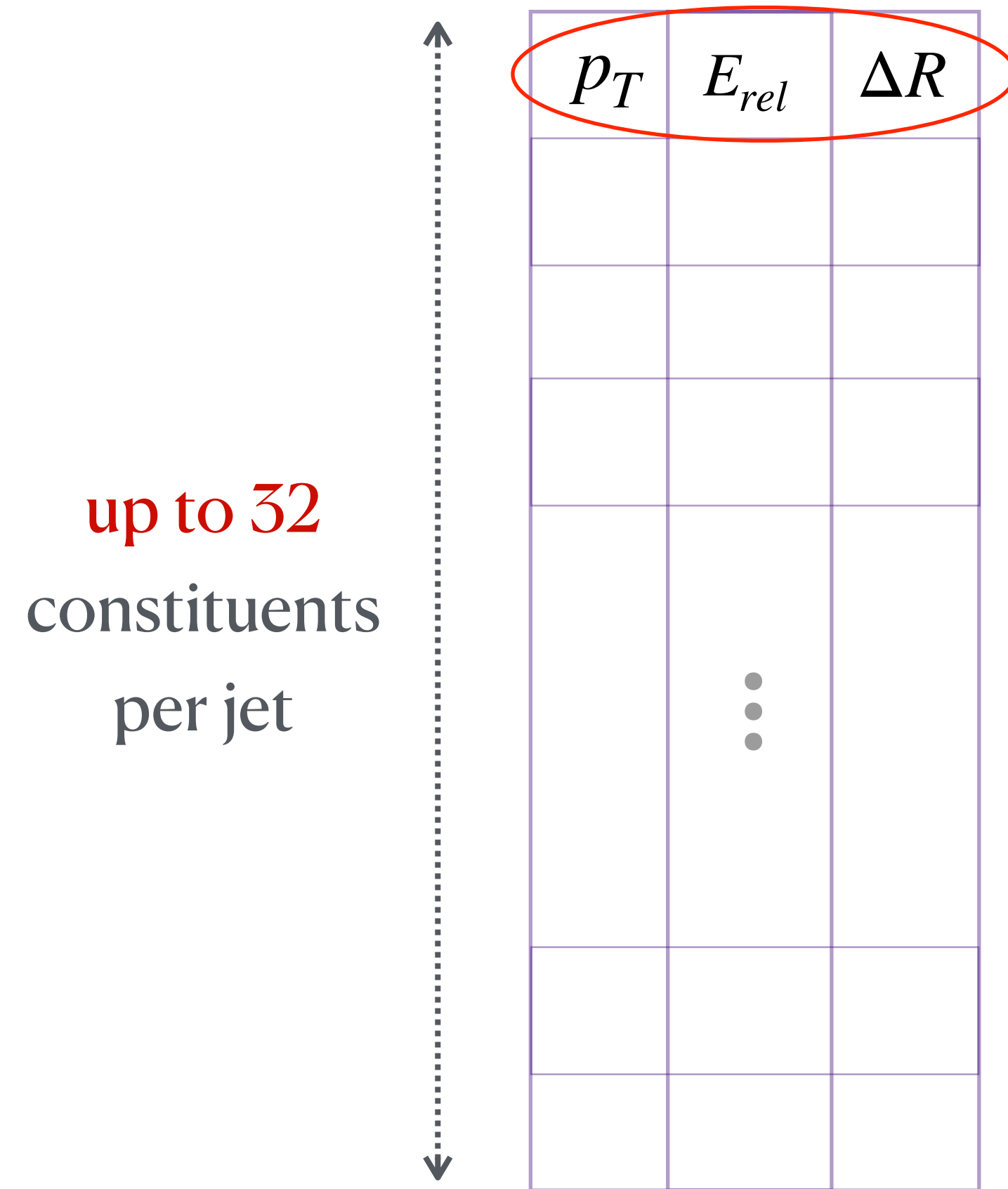


p_T	E_{rel}	ΔR
	⋮	

Data Embedding

(Reduced to most important features)

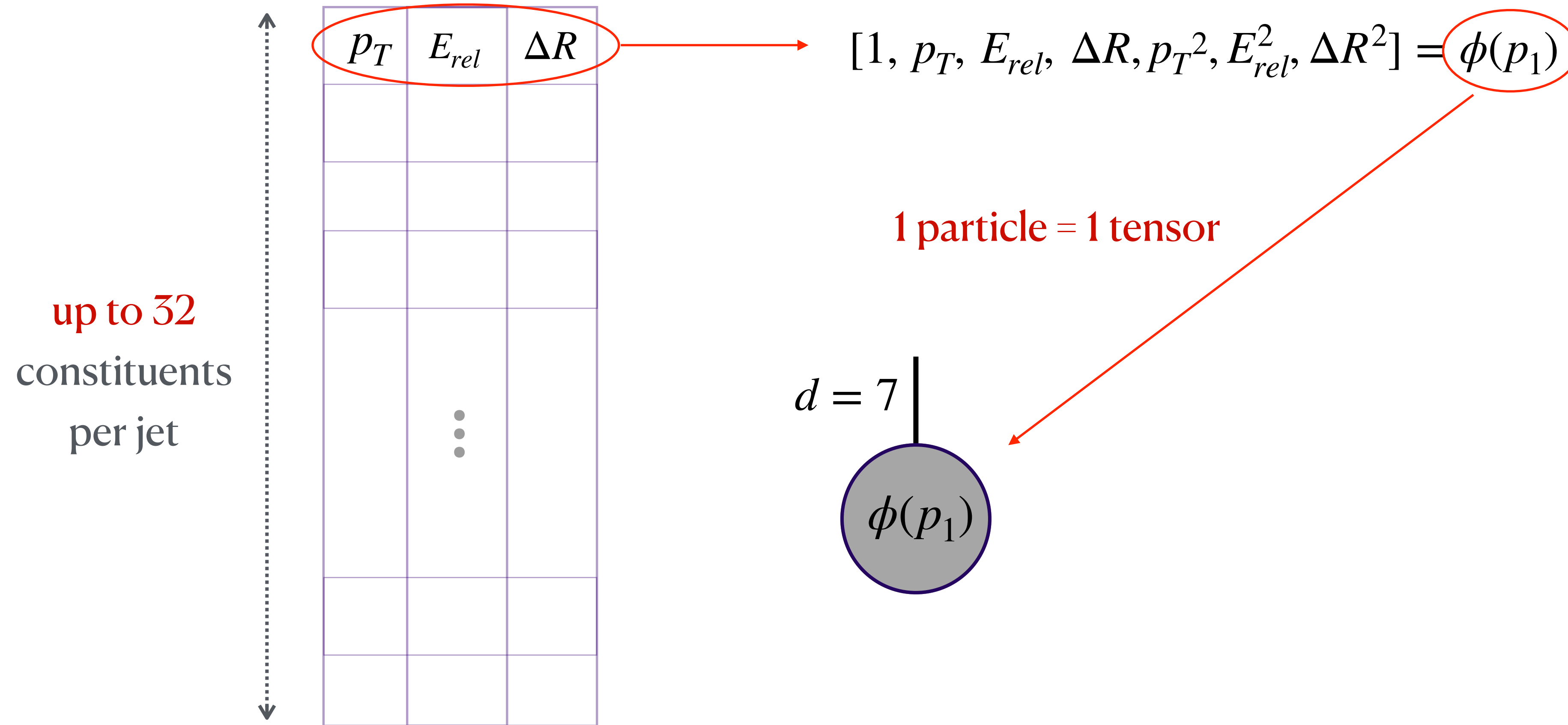
3 features



Data Embedding

(Reduced to most important features)

3 features



Data Embedding

(Reduced to most important features)

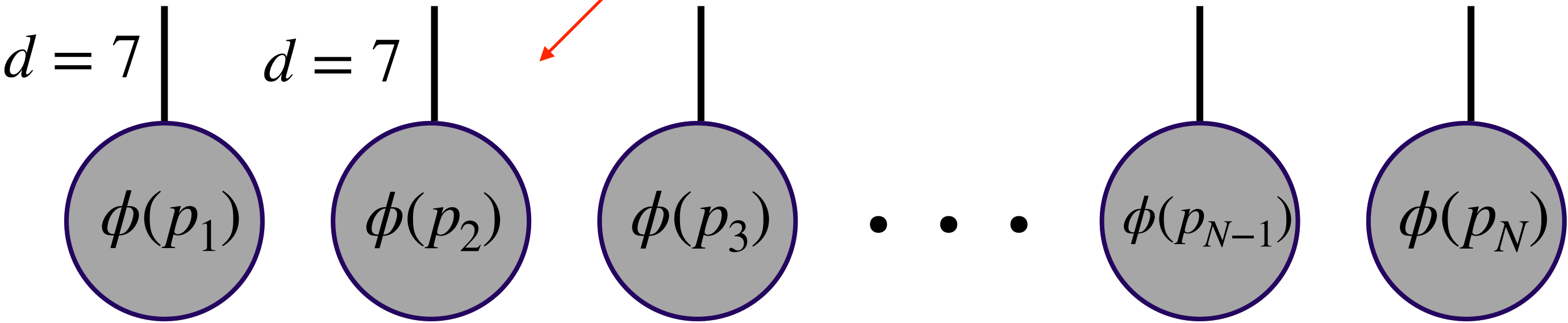
3 features

p_T	E_{rel}	ΔR
p_T	E_{rel}	ΔR
	$\vdots$	

up to 32
constituents
per jet

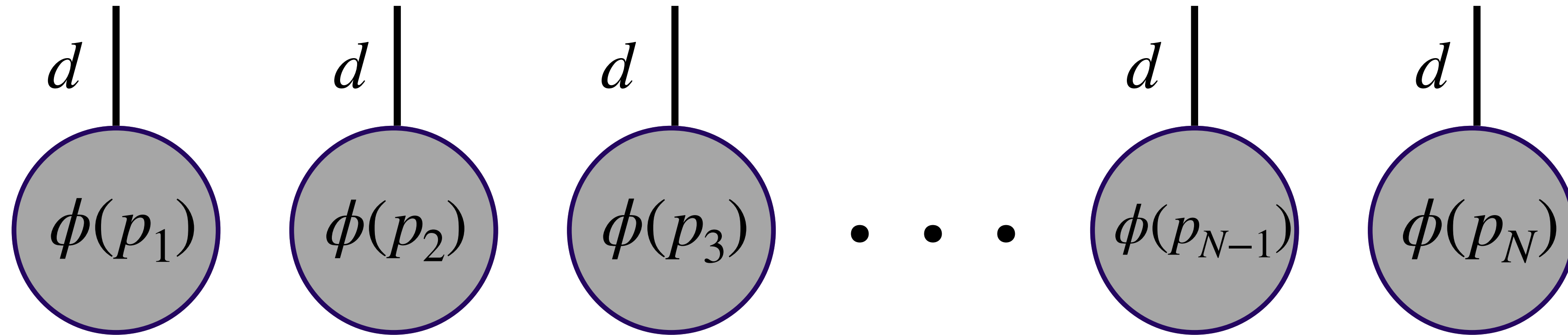
$[1, p_T, E_{rel}, \Delta R, p_T^2, E_{rel}^2, \Delta R^2] = \phi(p_2)$

1 particle = 1 tensor

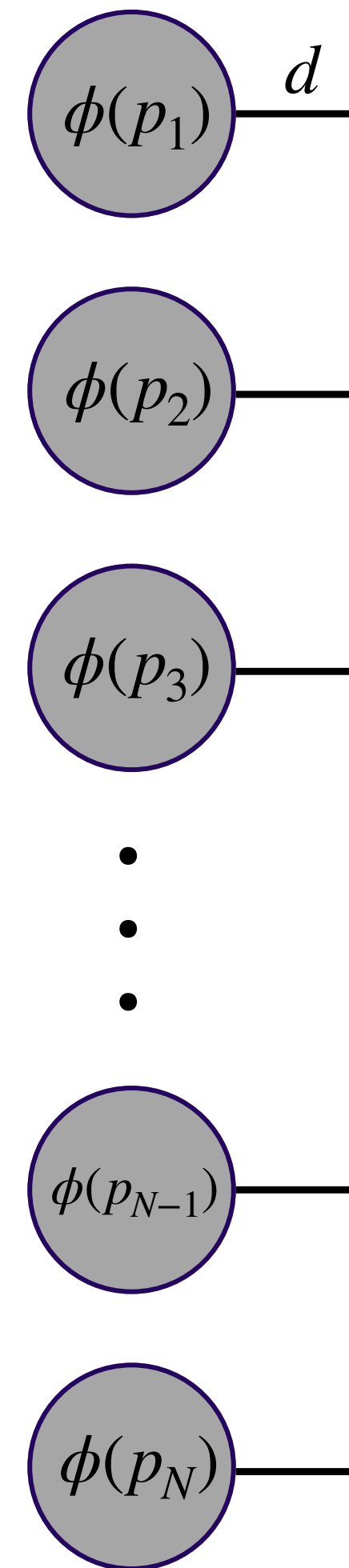


Data Embedding

1 particle = 1 tensor



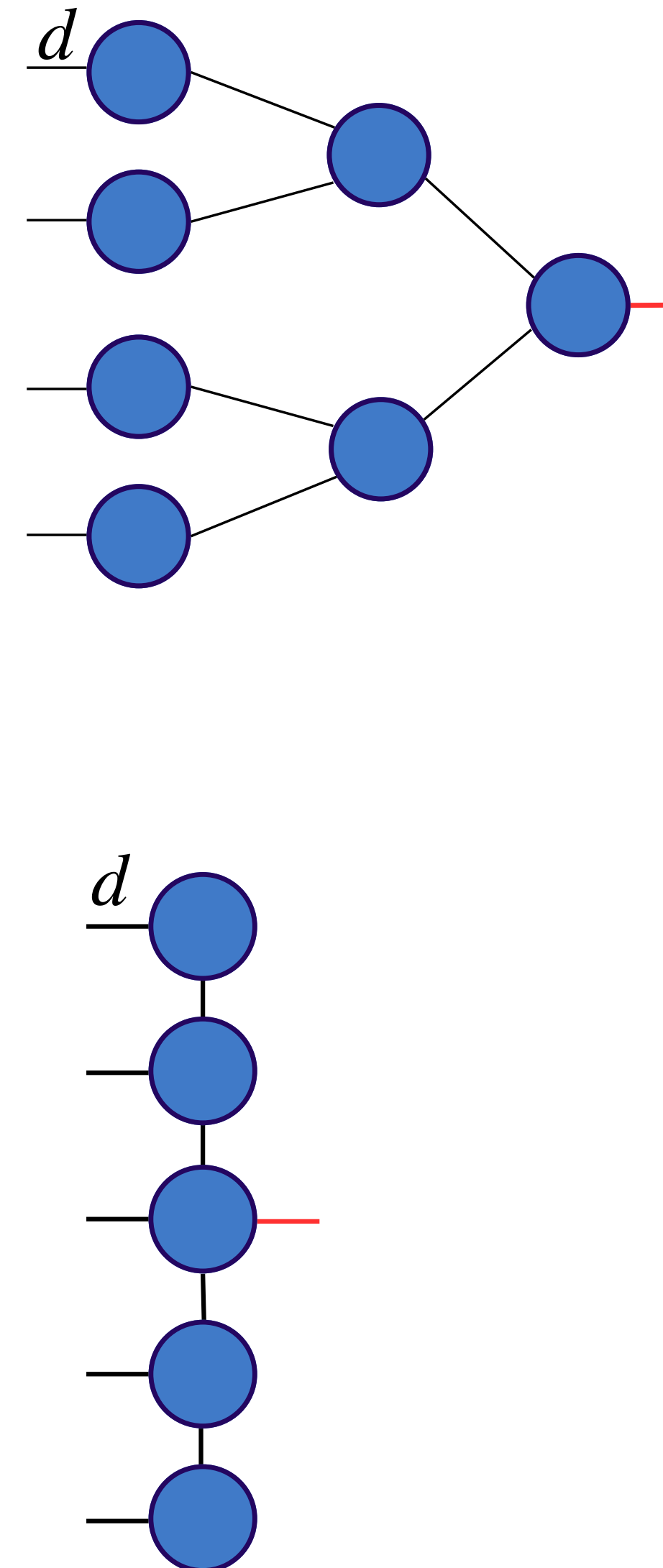
Embedding



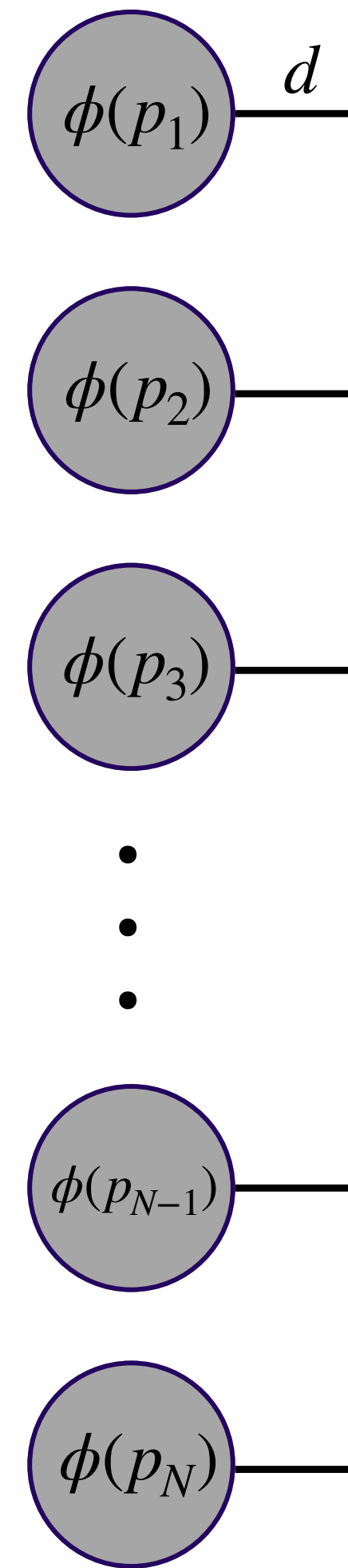
TN models

Contraction

+



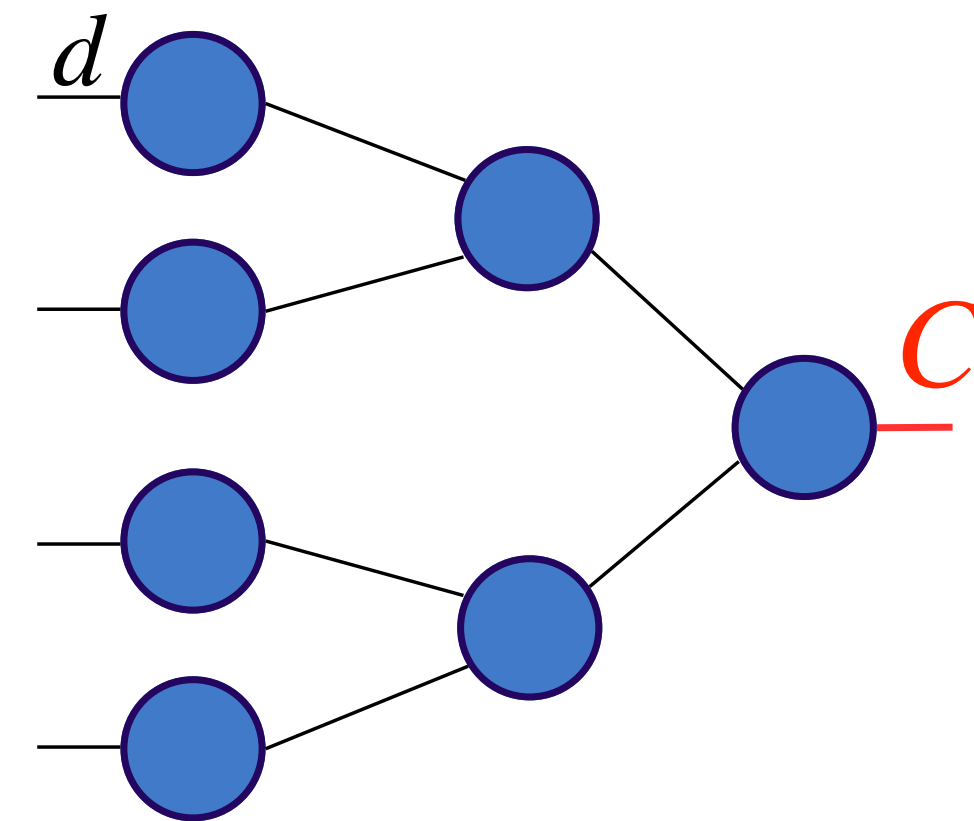
Embedding



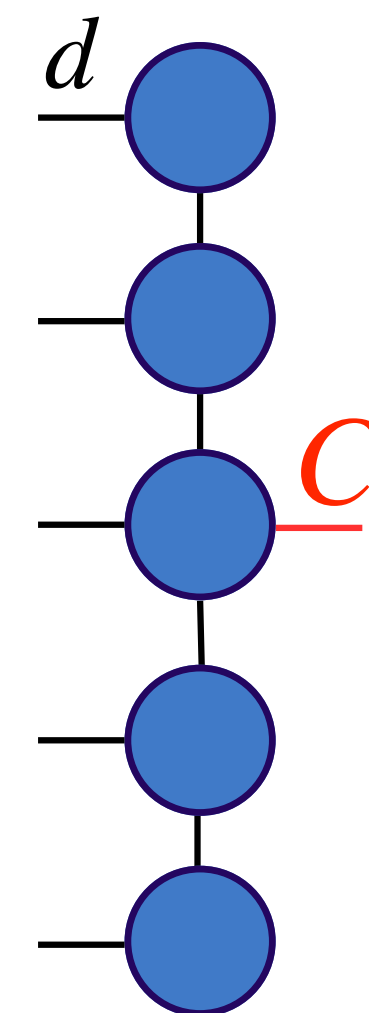
Contraction

+

TN models

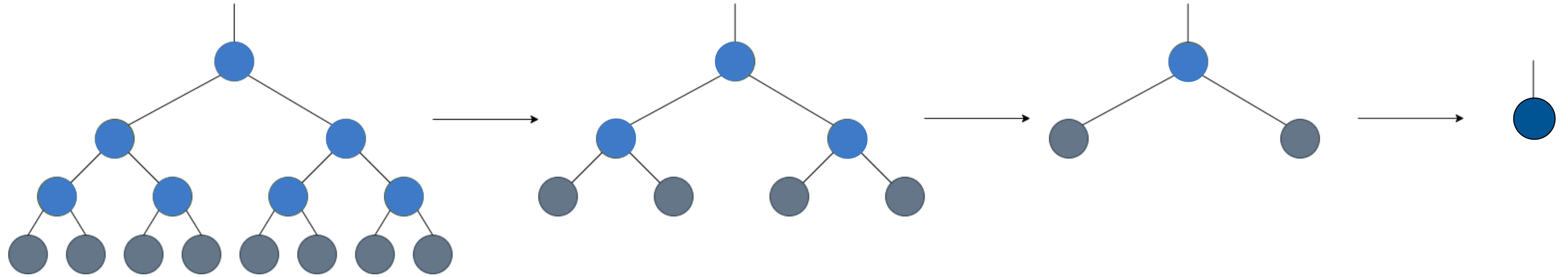


Vector of size C
as output



Contraction order

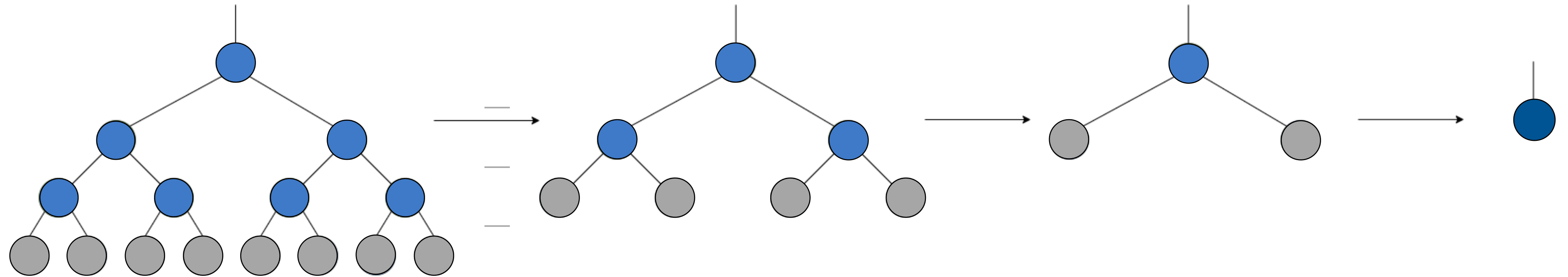
TTN



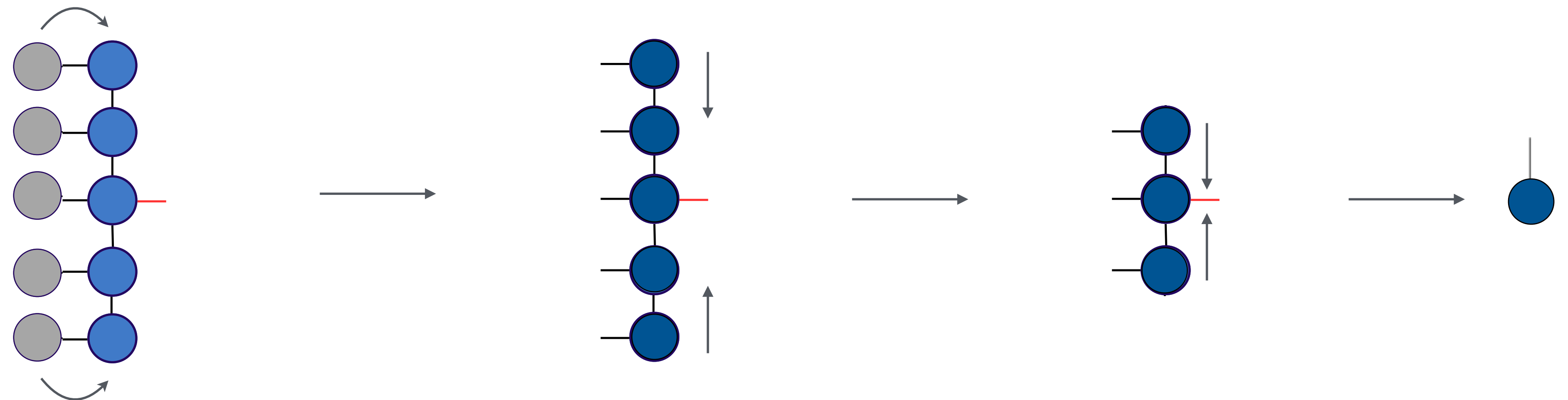
Contraction order

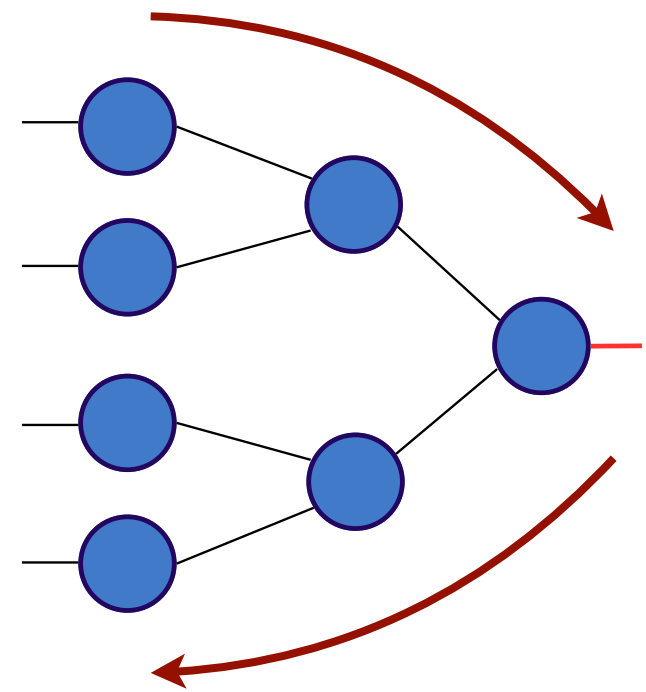
Fixed in the hardware implementation
- to avoid the search for optimal contraction path in runtime

TTN



MPS

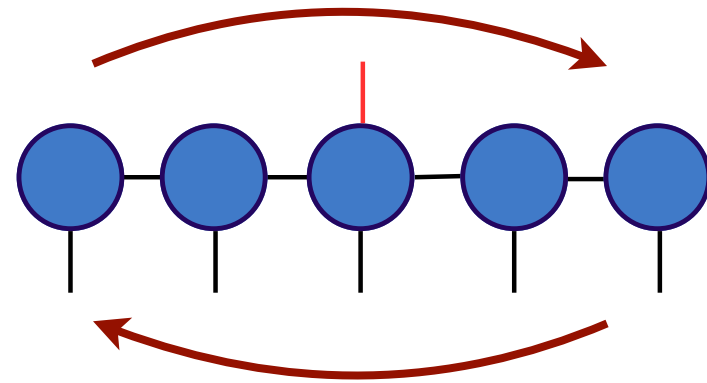




1. Floating- point optimization

Optimize both TN models with gradient-descent

- Includes hyperparameter search
 - bond dimension, tensor initialization function, loss function



2. Post-training quantization

Reduce precision of the models after training to fit into L1 trigger restrictions

3. FPGA Deployment

Deploy models on FPGA and perform analysis of resource usage and latency in inference time

FPGA implementation

TTN

Firmware in VHDL (Vivado)

Latency ($\mathcal{O}(\log_2 B^2)$), resource usage ($\mathcal{O}(B^3)$)

Optimized parallelism - low inference latency

Maxes out available resources for performance

Feature map in Look-Up Tables (LUTs)

MPS

HLS (Vitis)

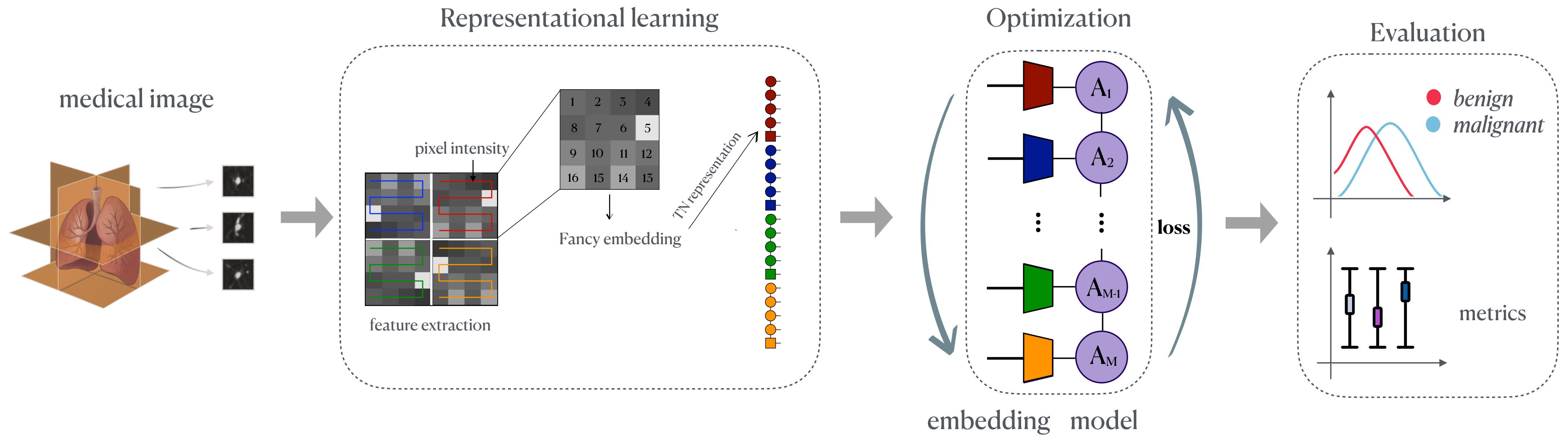
Latency VS resource usage can be tuned in HLS

Faster development and testing cycles (C++ $\rightarrow$ FPGA)

Balances performance with available resources

Weights implemented in LUTs

Tensor Networks for Cancer Detection in Medical Imaging



Thank you!